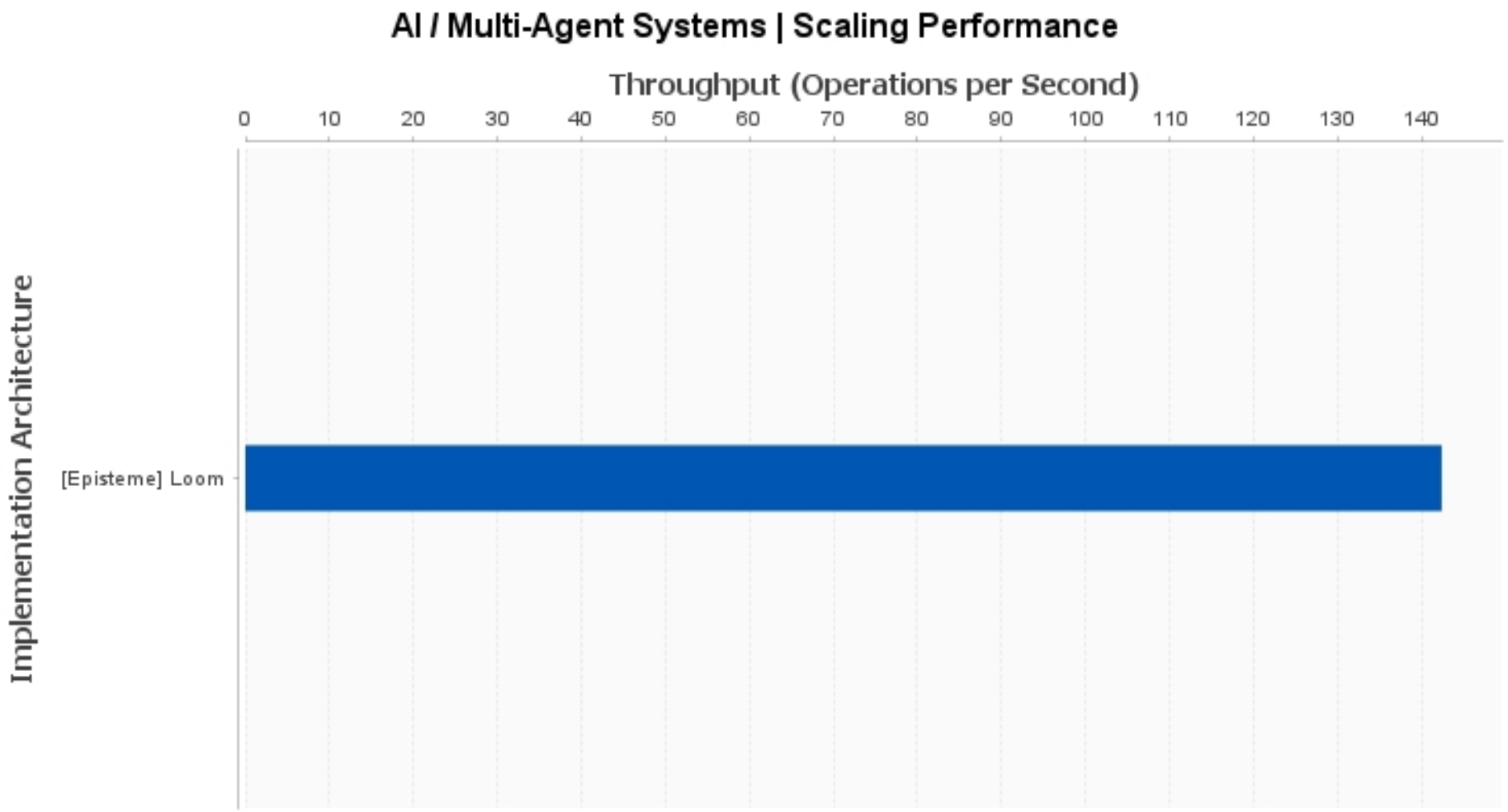
Episteme Benchmark Executive Summary

Precision Performance Analytics | Generated: 2026-03-04 22:53:45

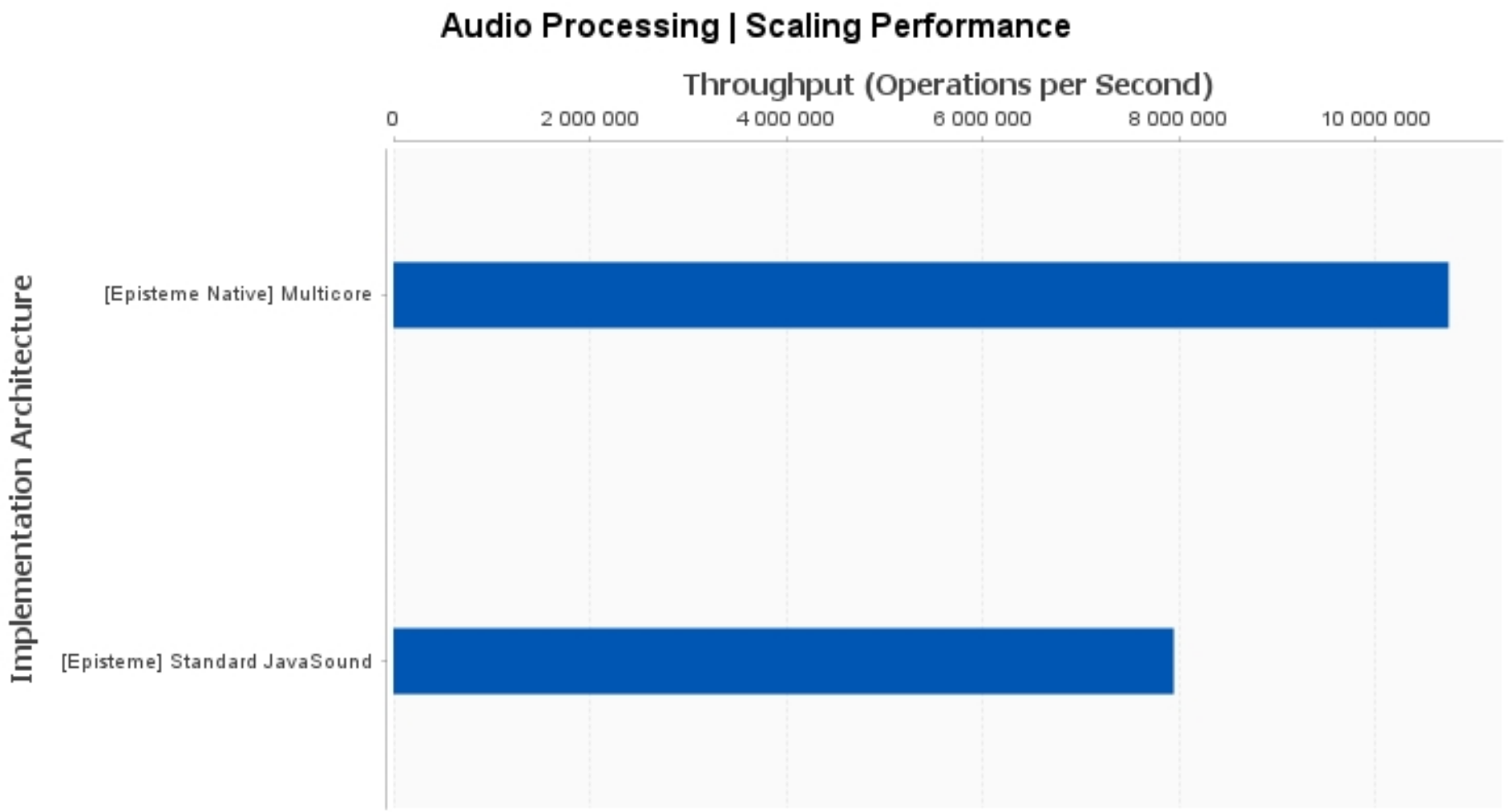
Operating System	Windows 11 (amd64)
Java Runtime	25 (Oracle Corporation)
Logical Processors	4

AI / Multi-Agent Systems

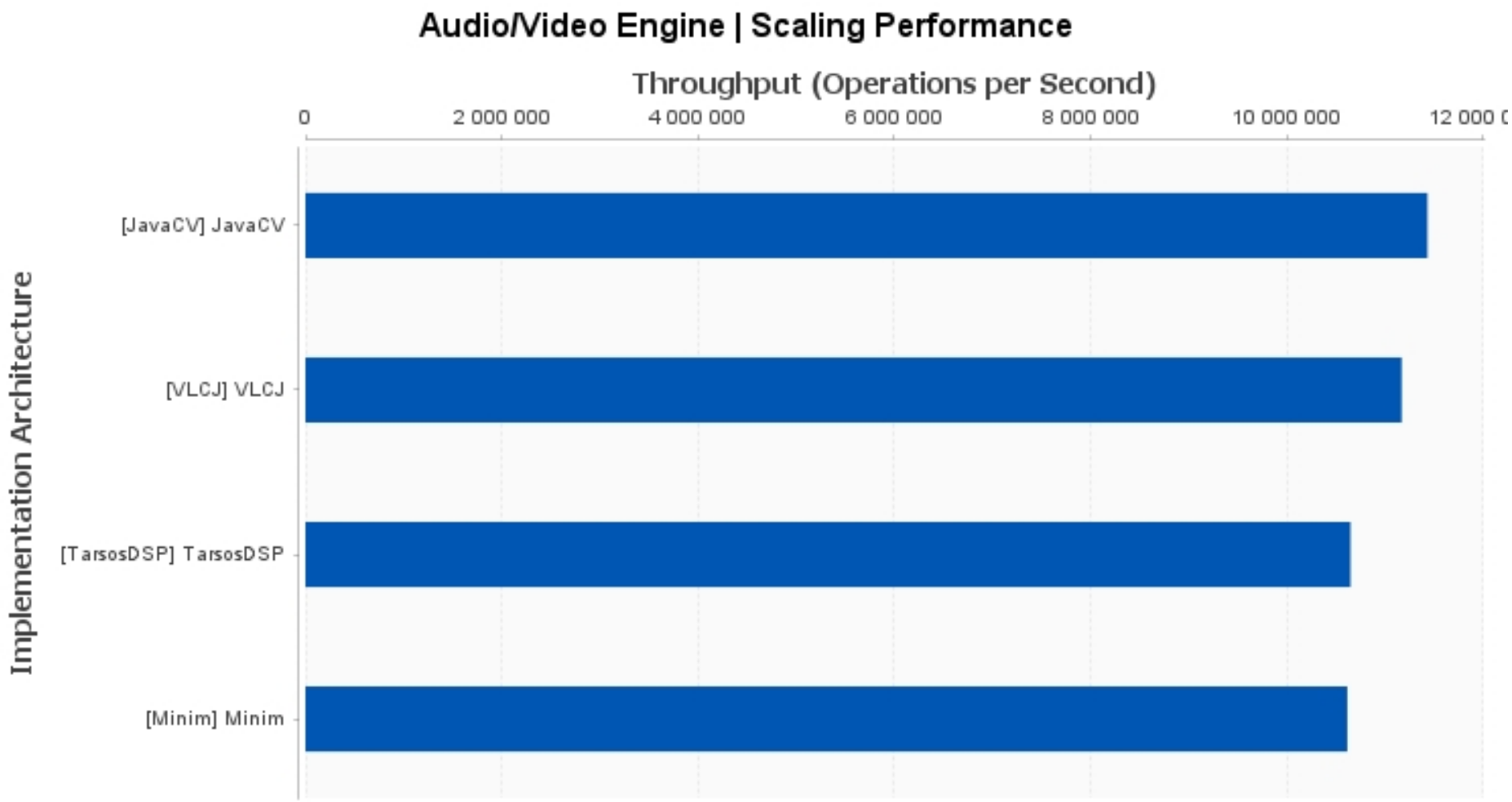
Audio Processing



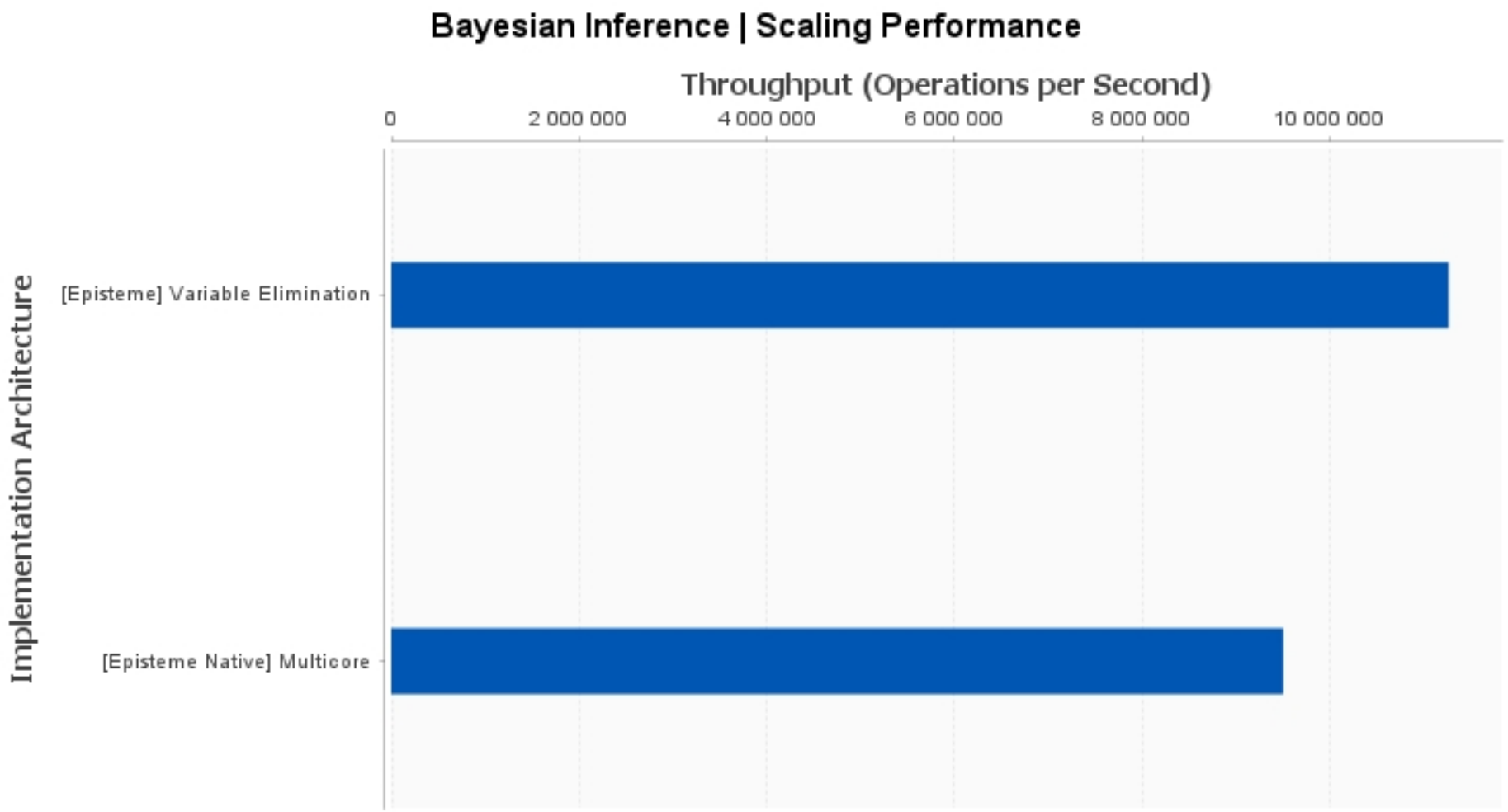
Audio/Video Engine



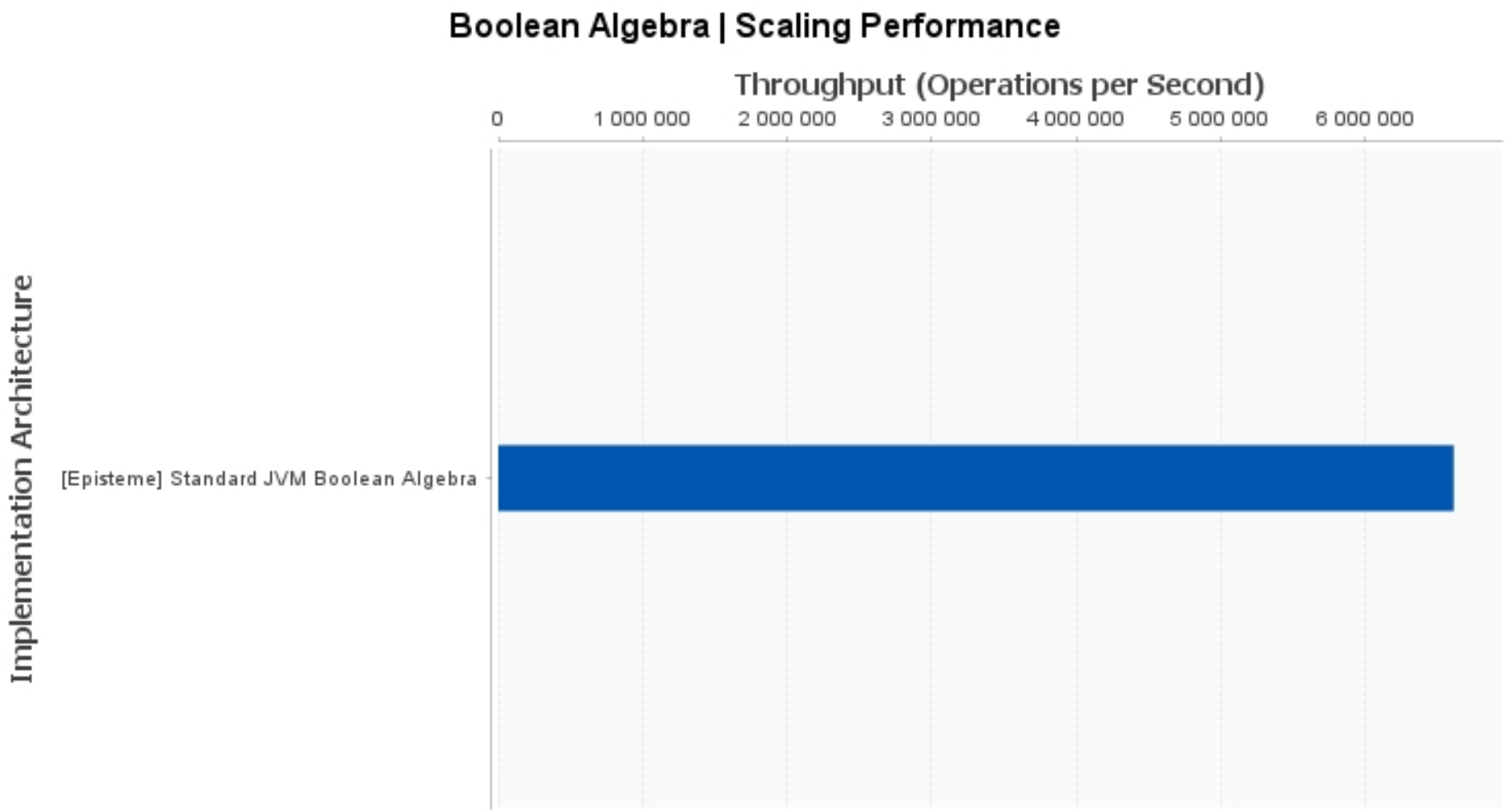
Bayesian Inference



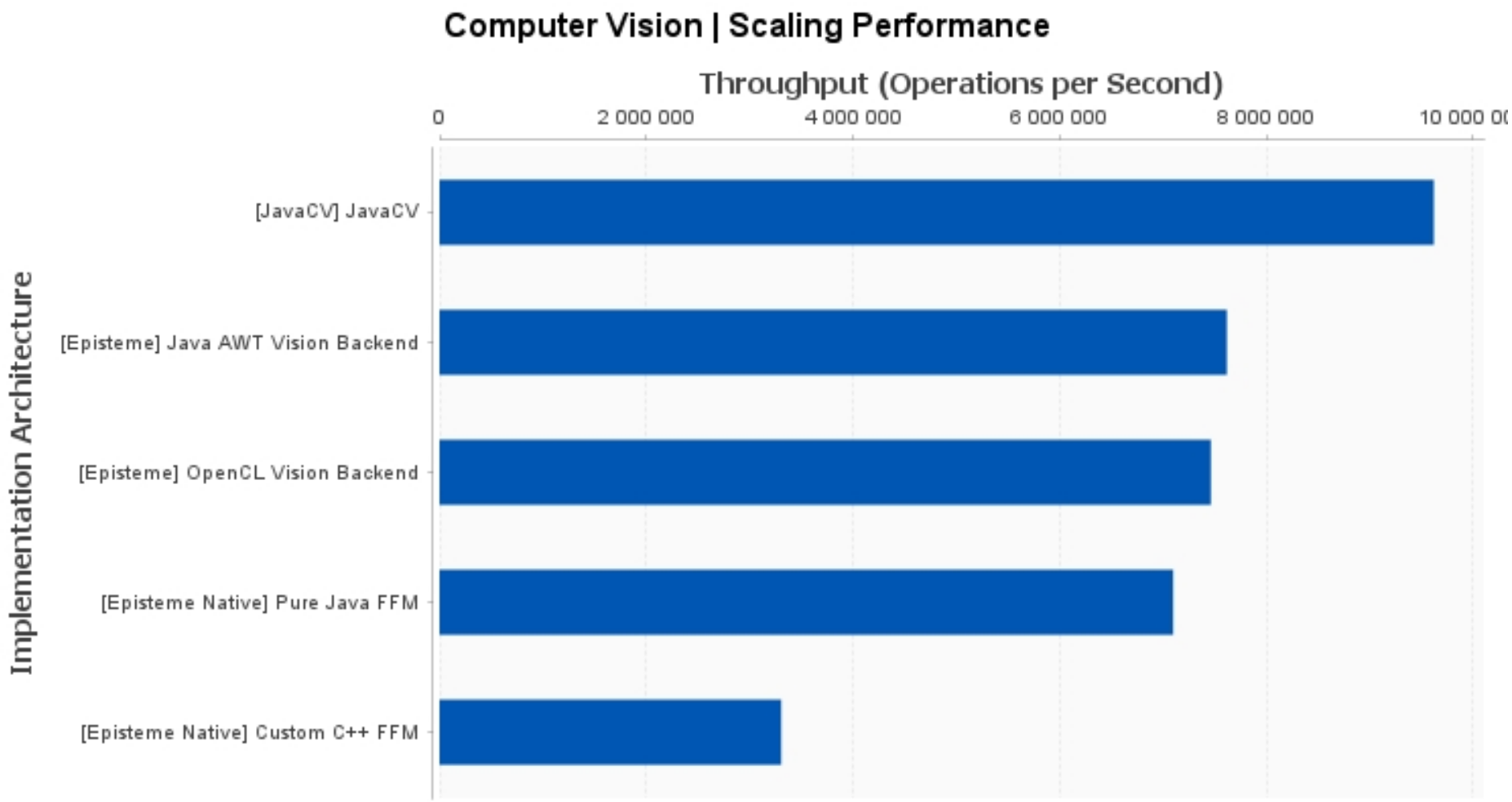
Boolean Algebra



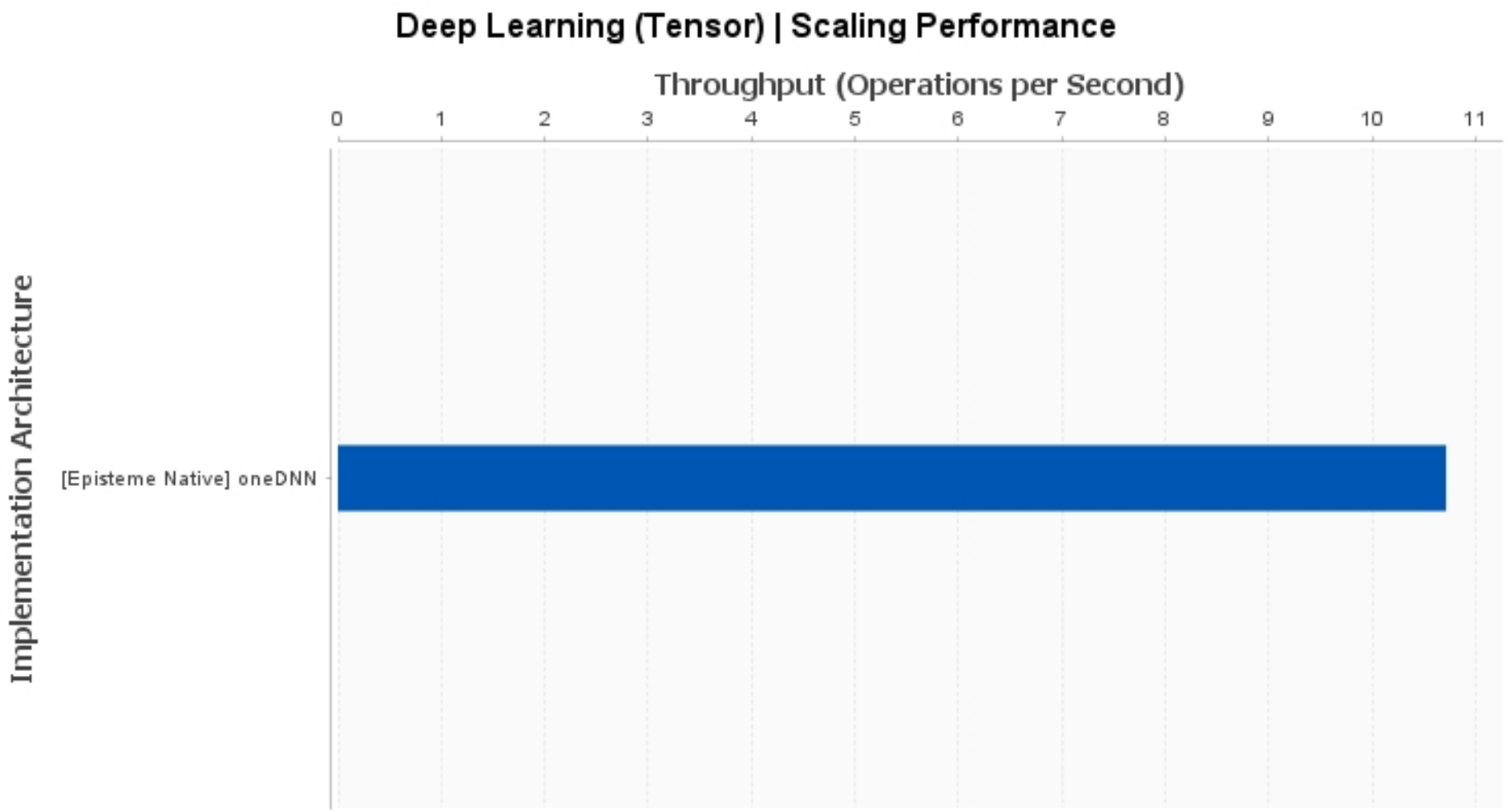
Computer Vision



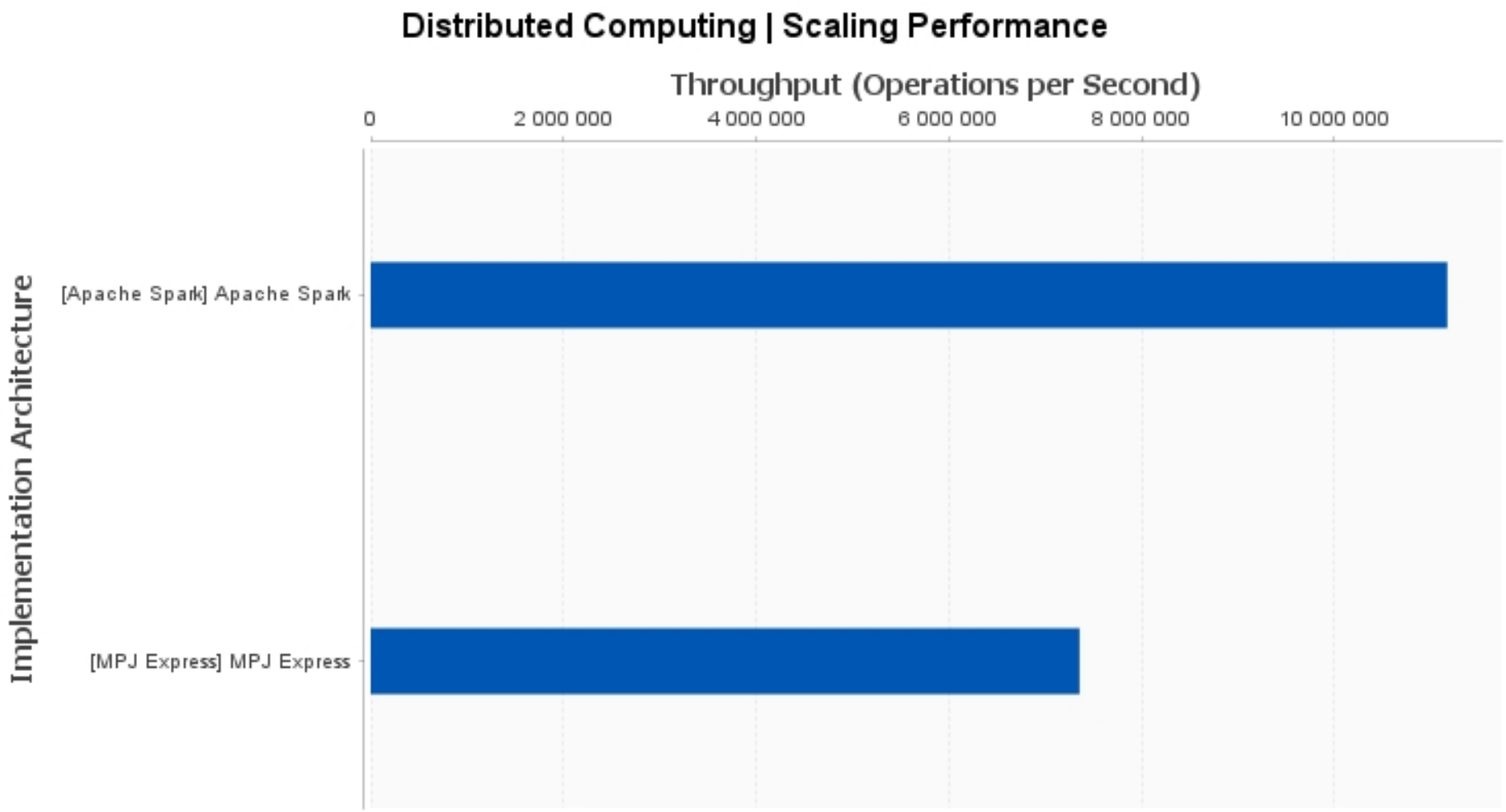
Deep Learning (Tensor)



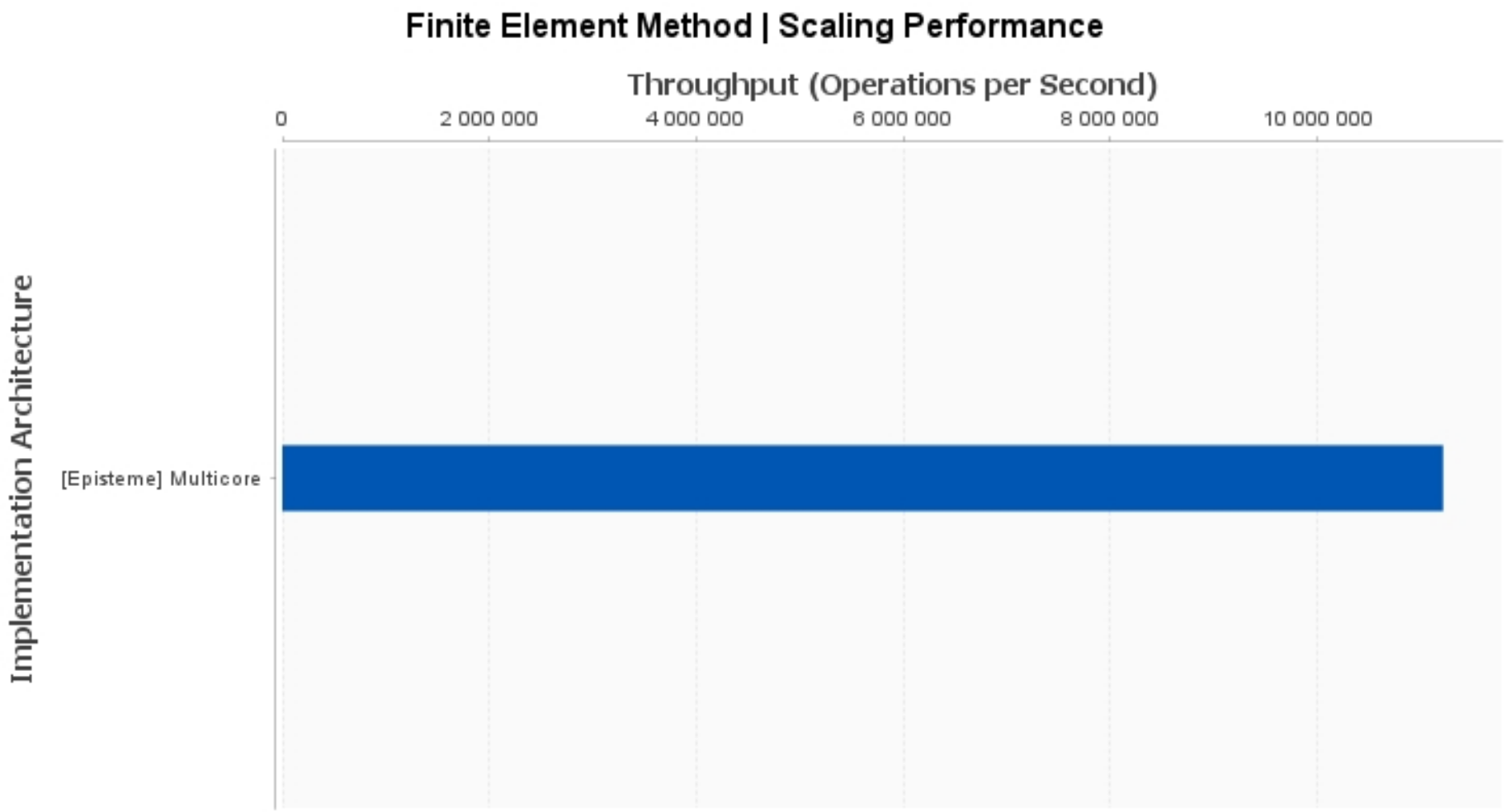
Distributed Computing



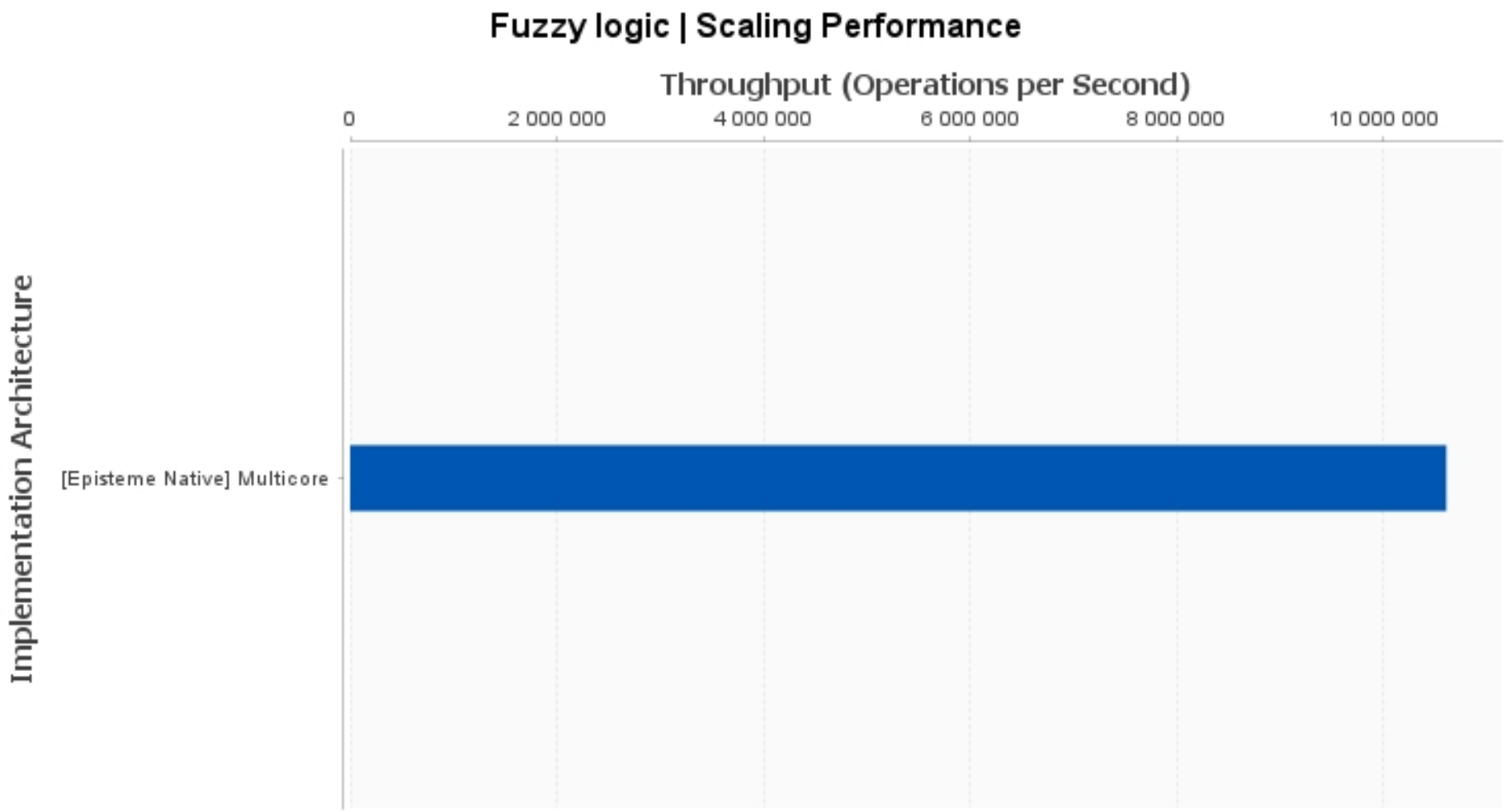
Finite Element Method


Distributed Computing | Scaling Performance
Throughput (Operations per Second)
0
2 000 000
4 000 000
6 000 000
8 000 000
10 000 000
Implementation Architecture
[Apache Spark] Apache Spark
[MPJ Express] MPJ Express

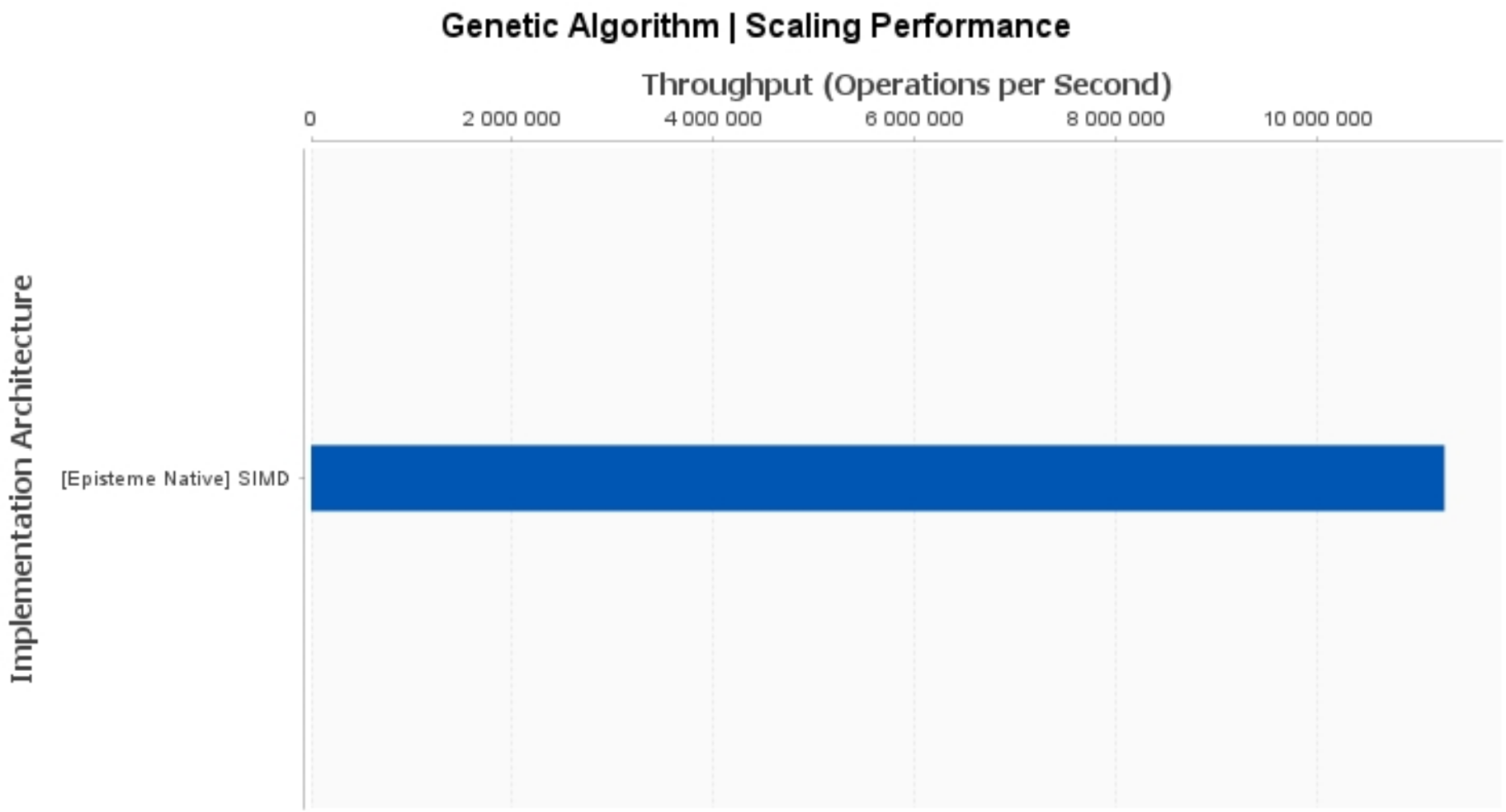
Fuzzy logic



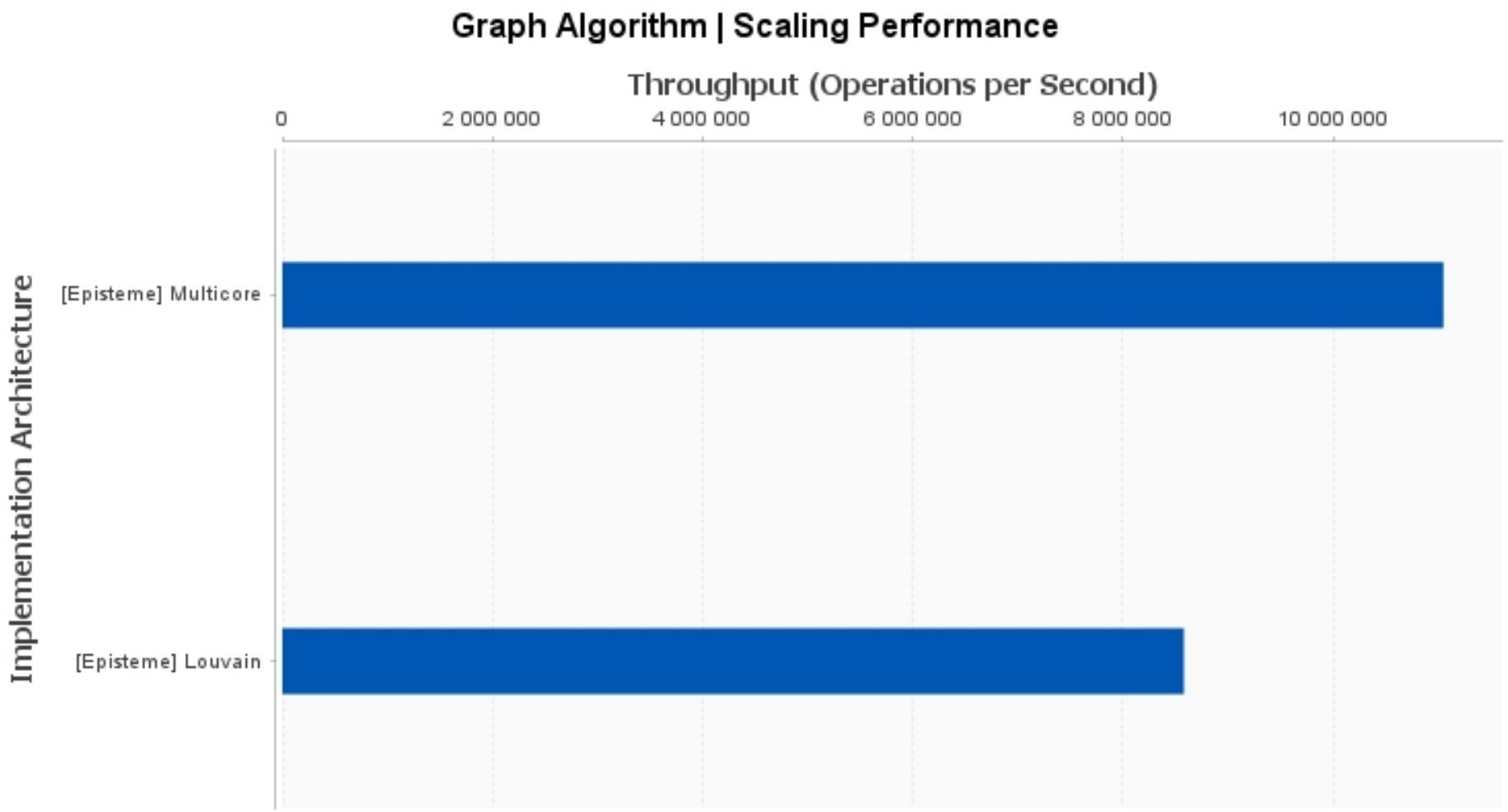
Genetic Algorithm



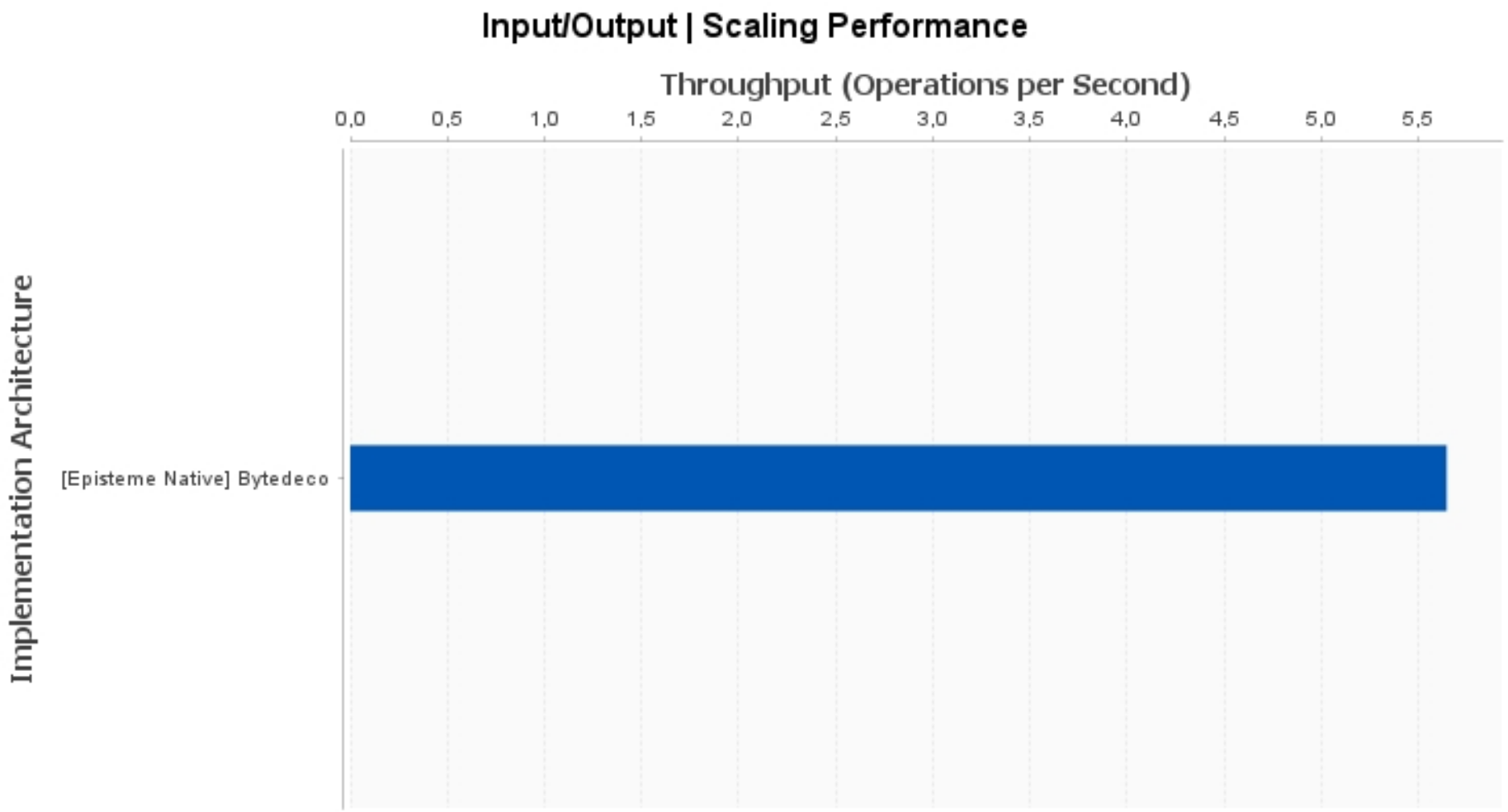
Graph Algorithm



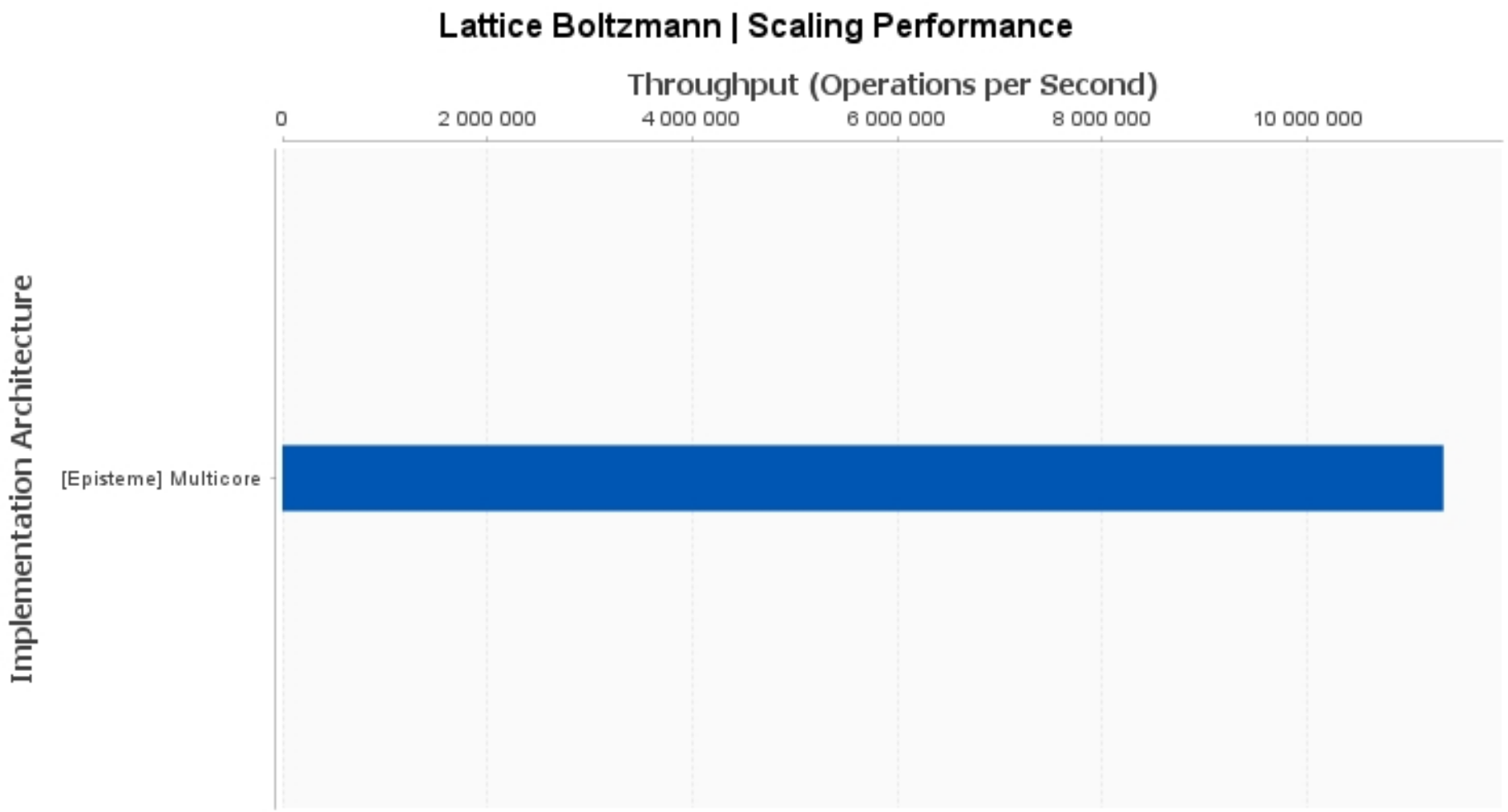
Input/Output



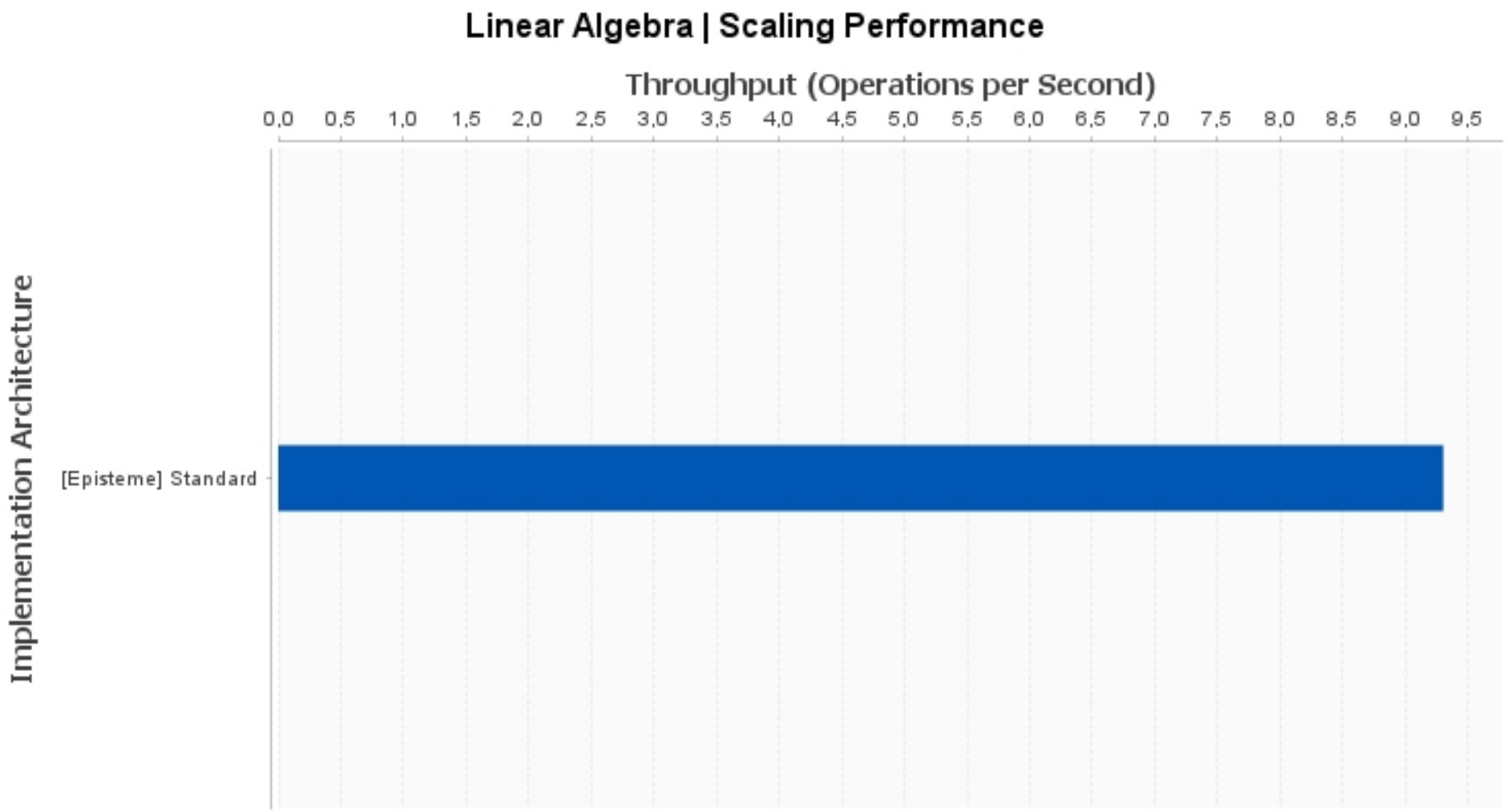
Lattice Boltzmann



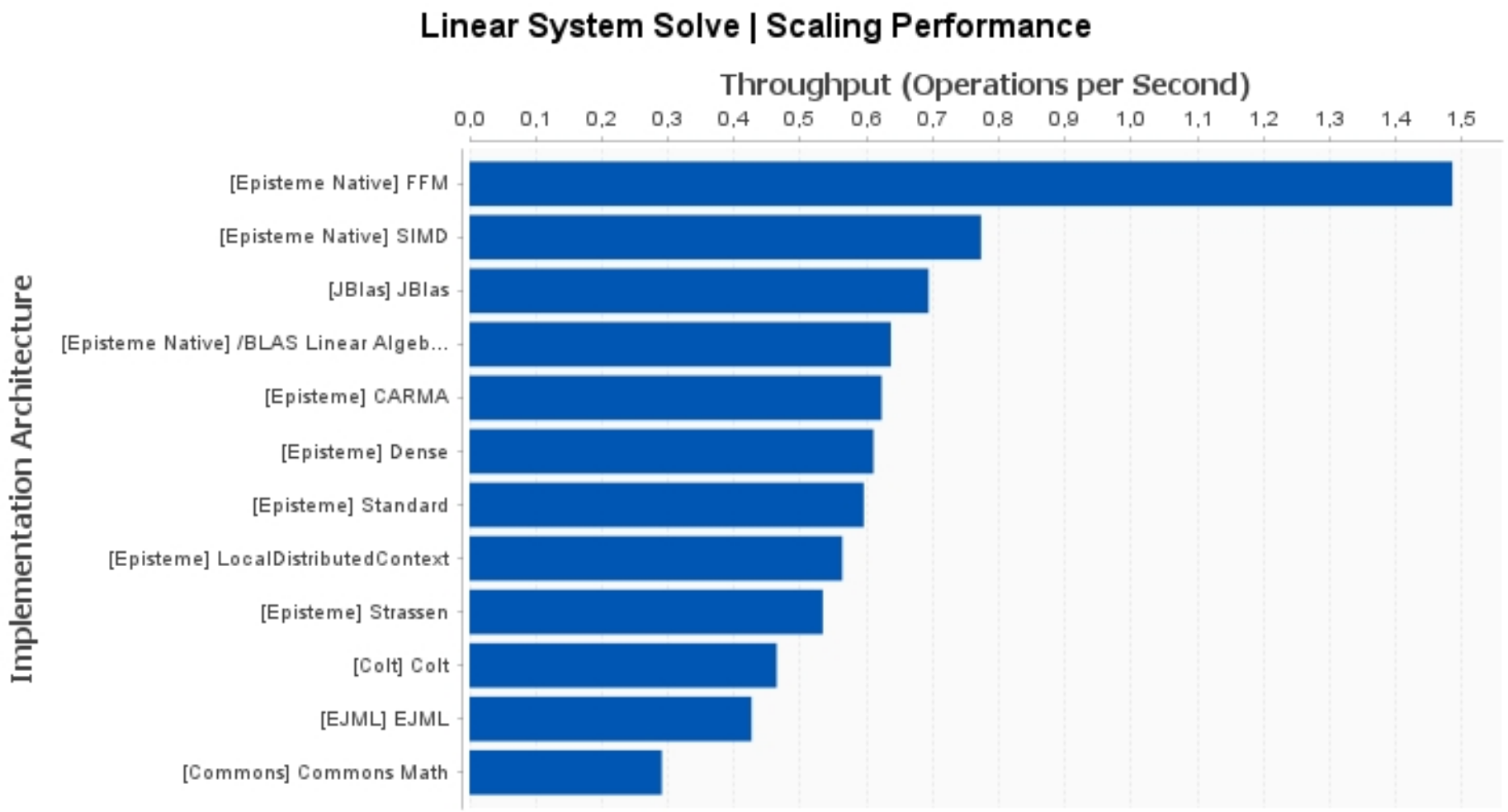
Linear Algebra



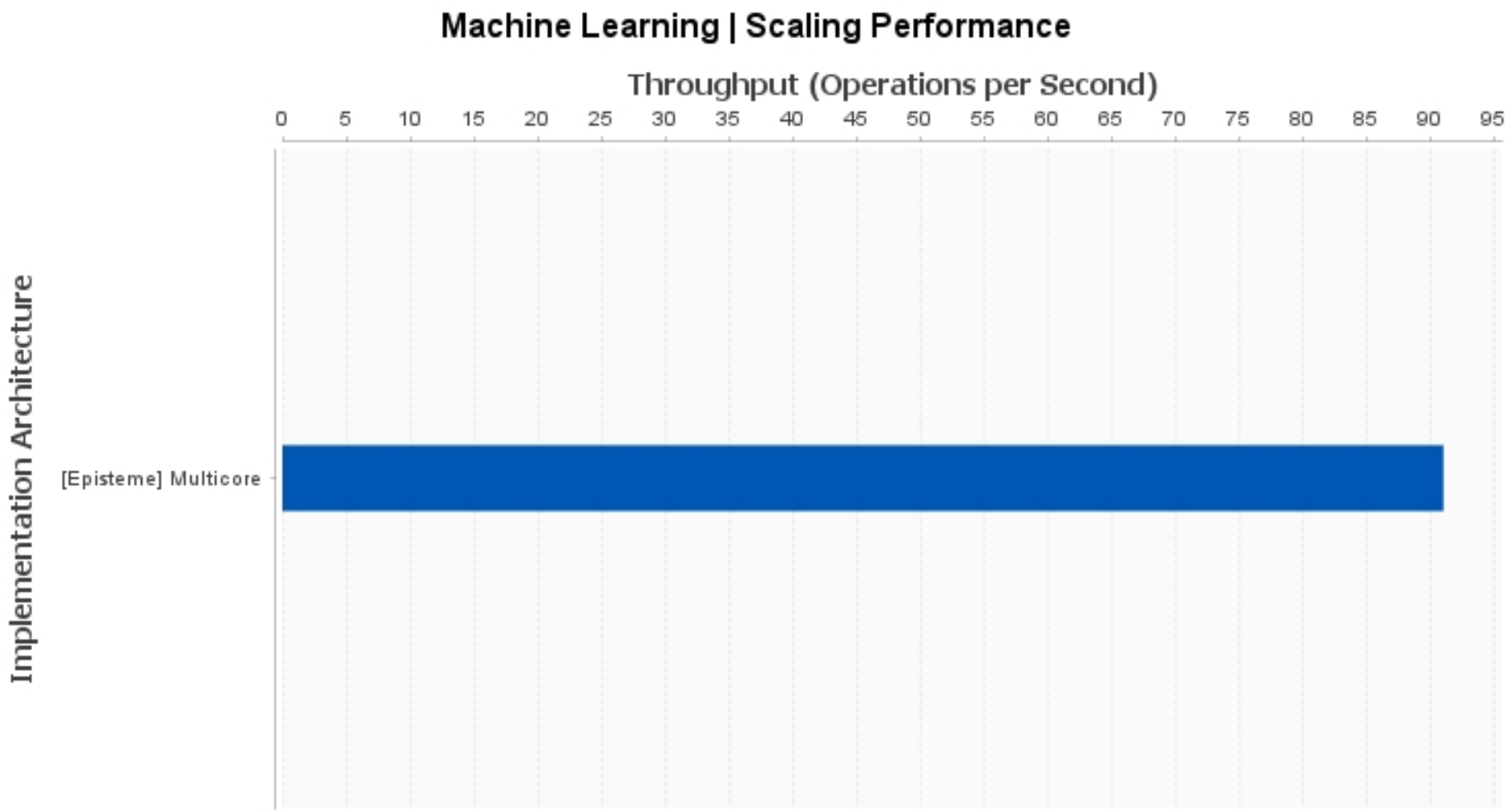
Linear System Solve



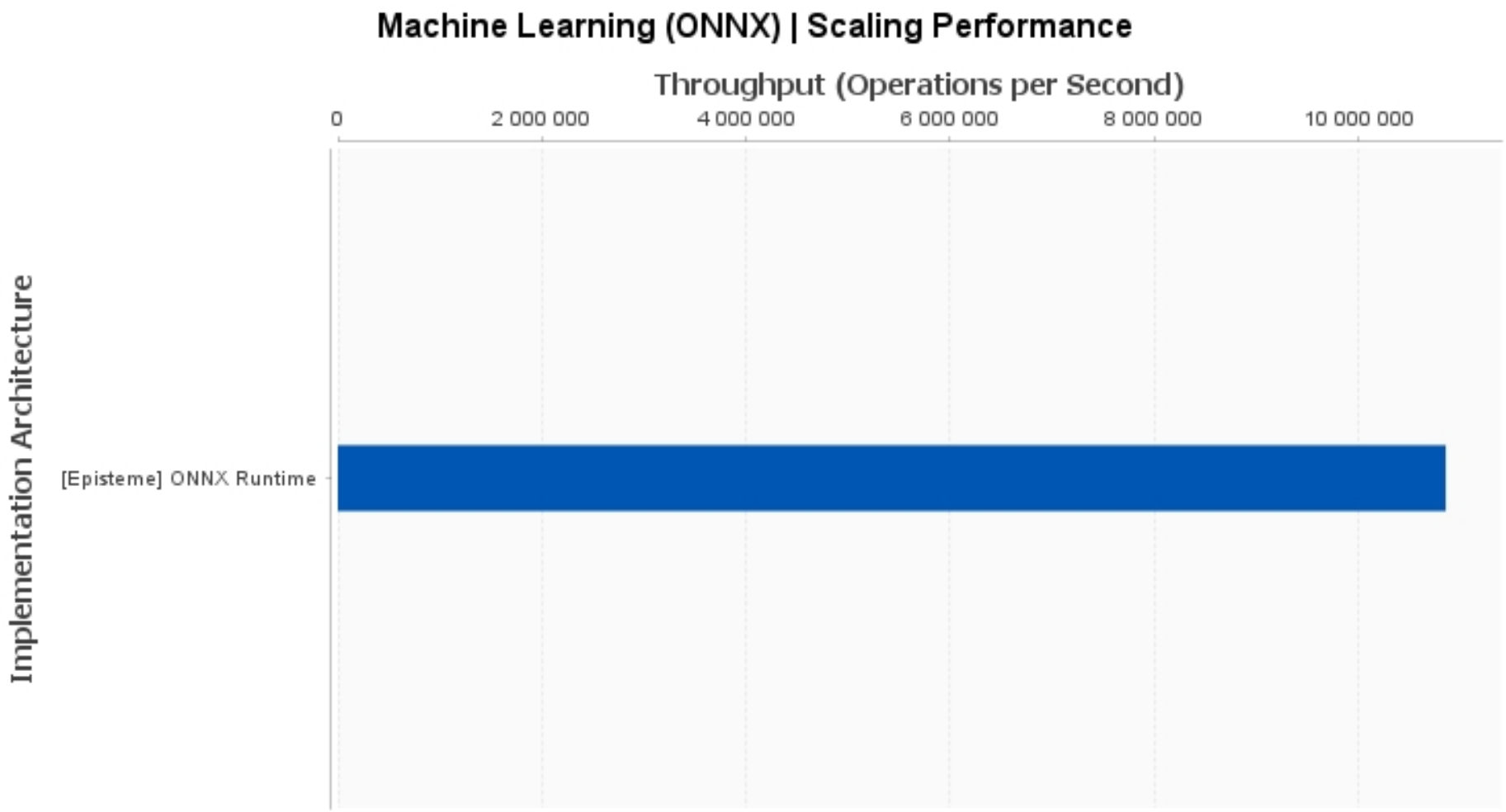
Machine Learning



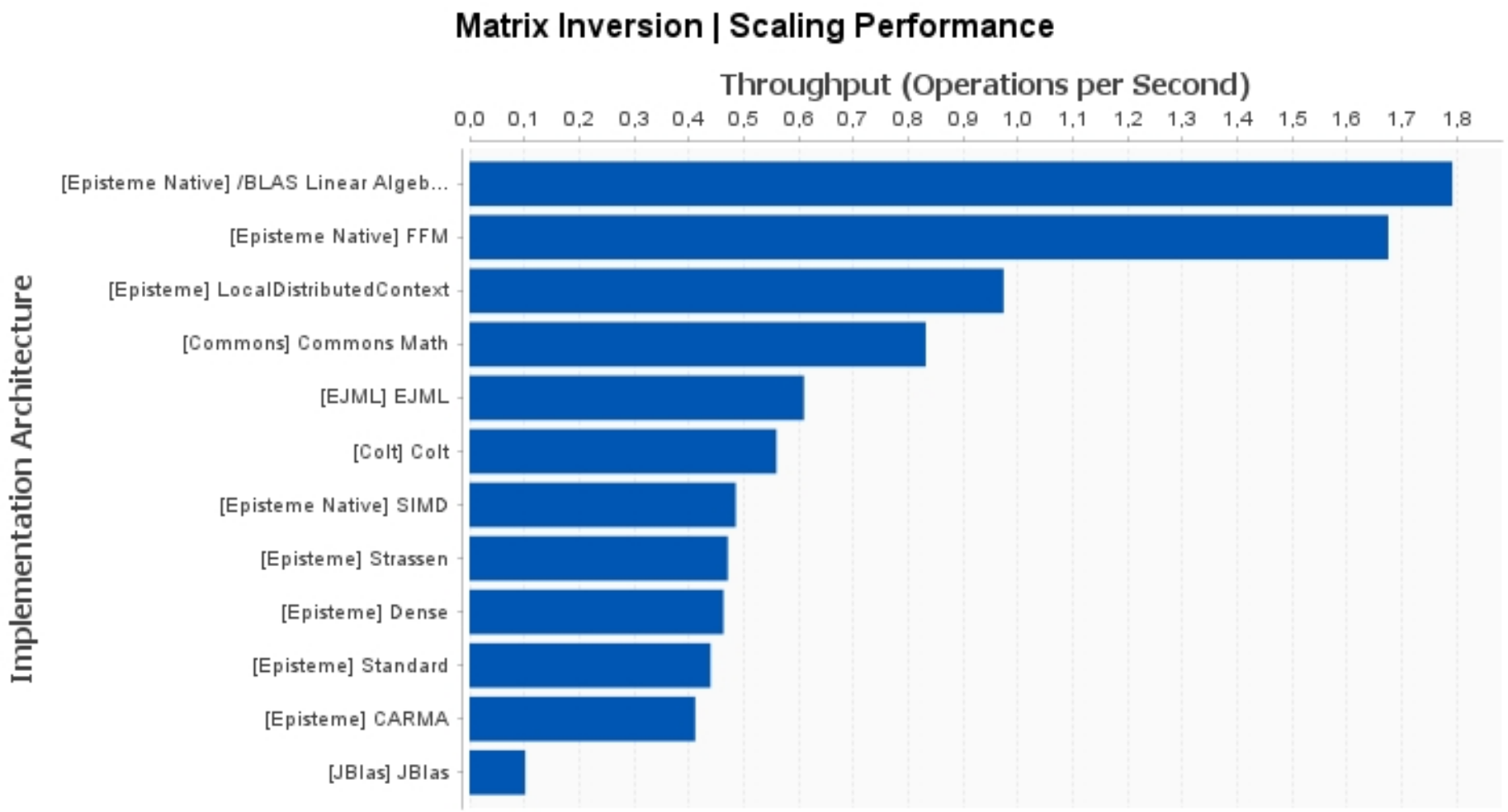
Machine Learning (ONNX)



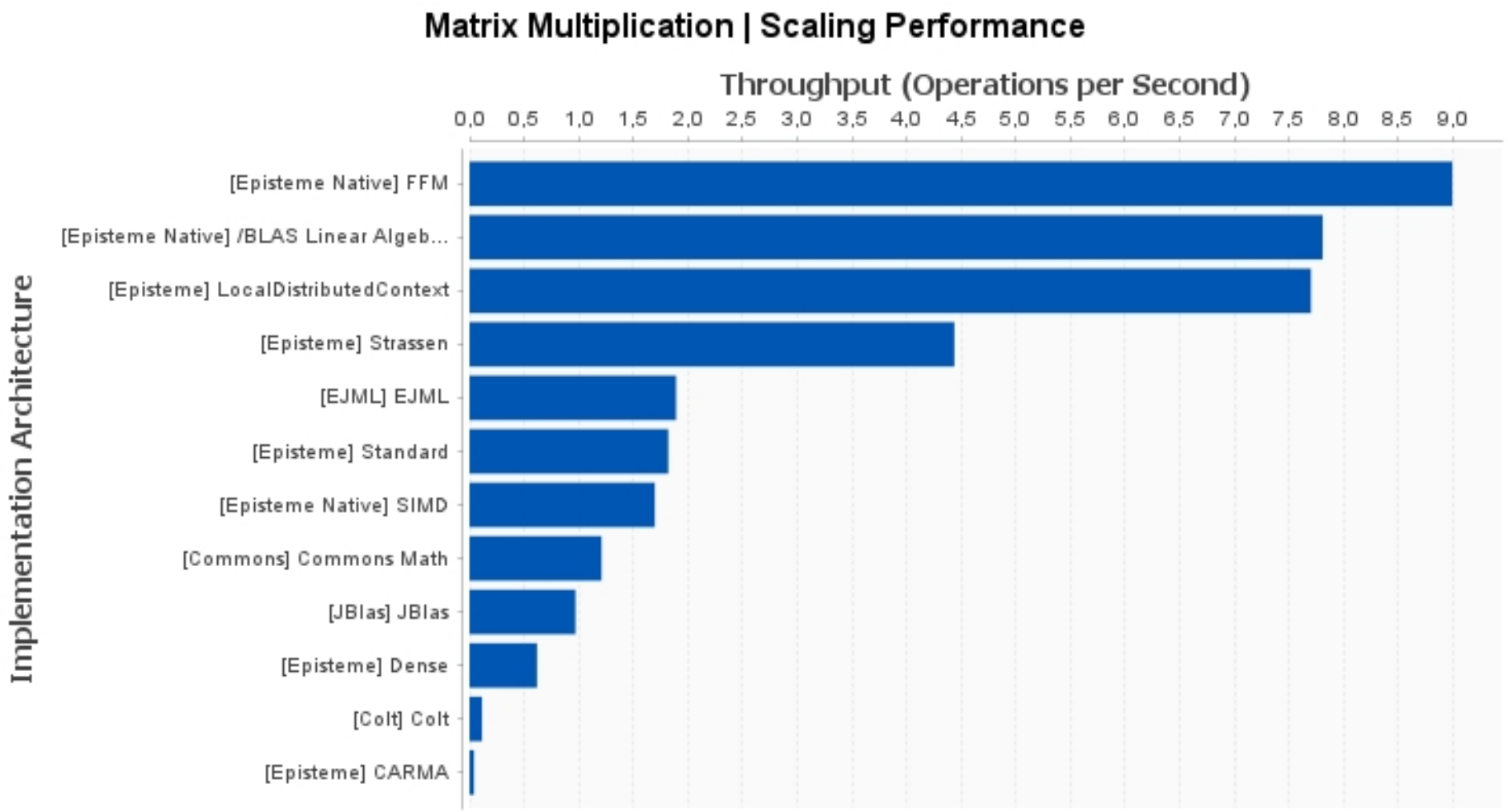
Matrix Inversion



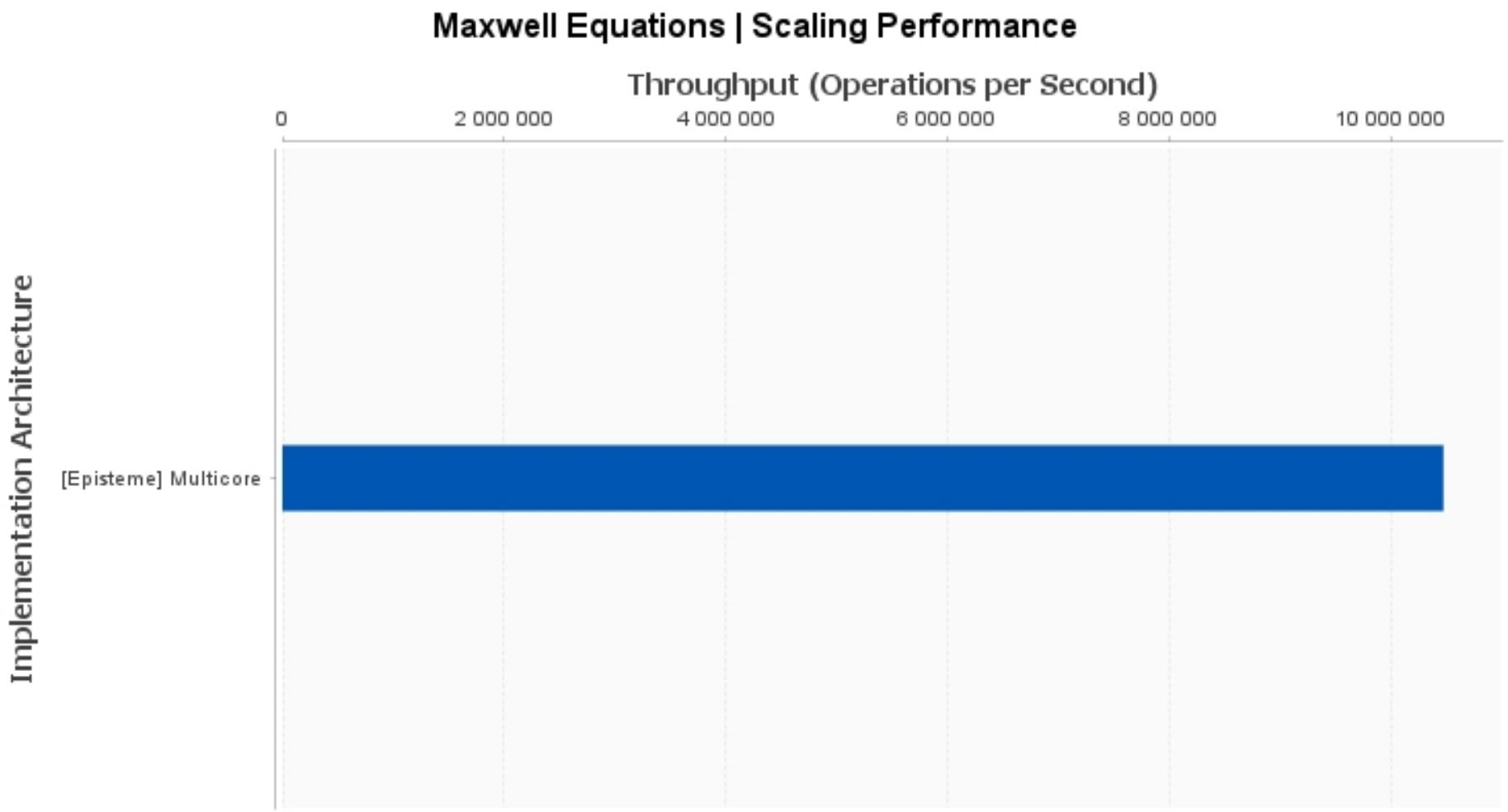
Matrix Multiplication



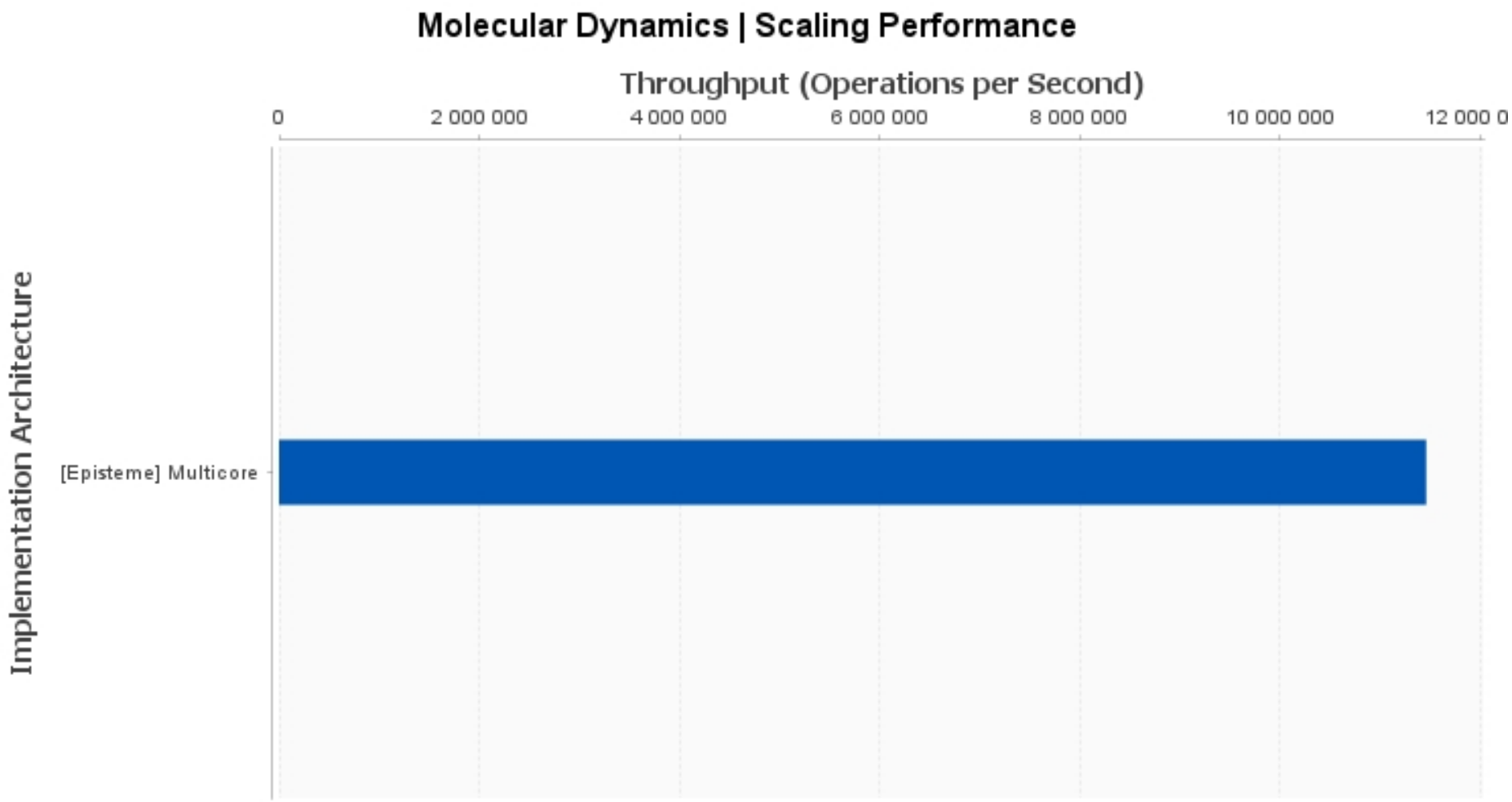
Maxwell Equations



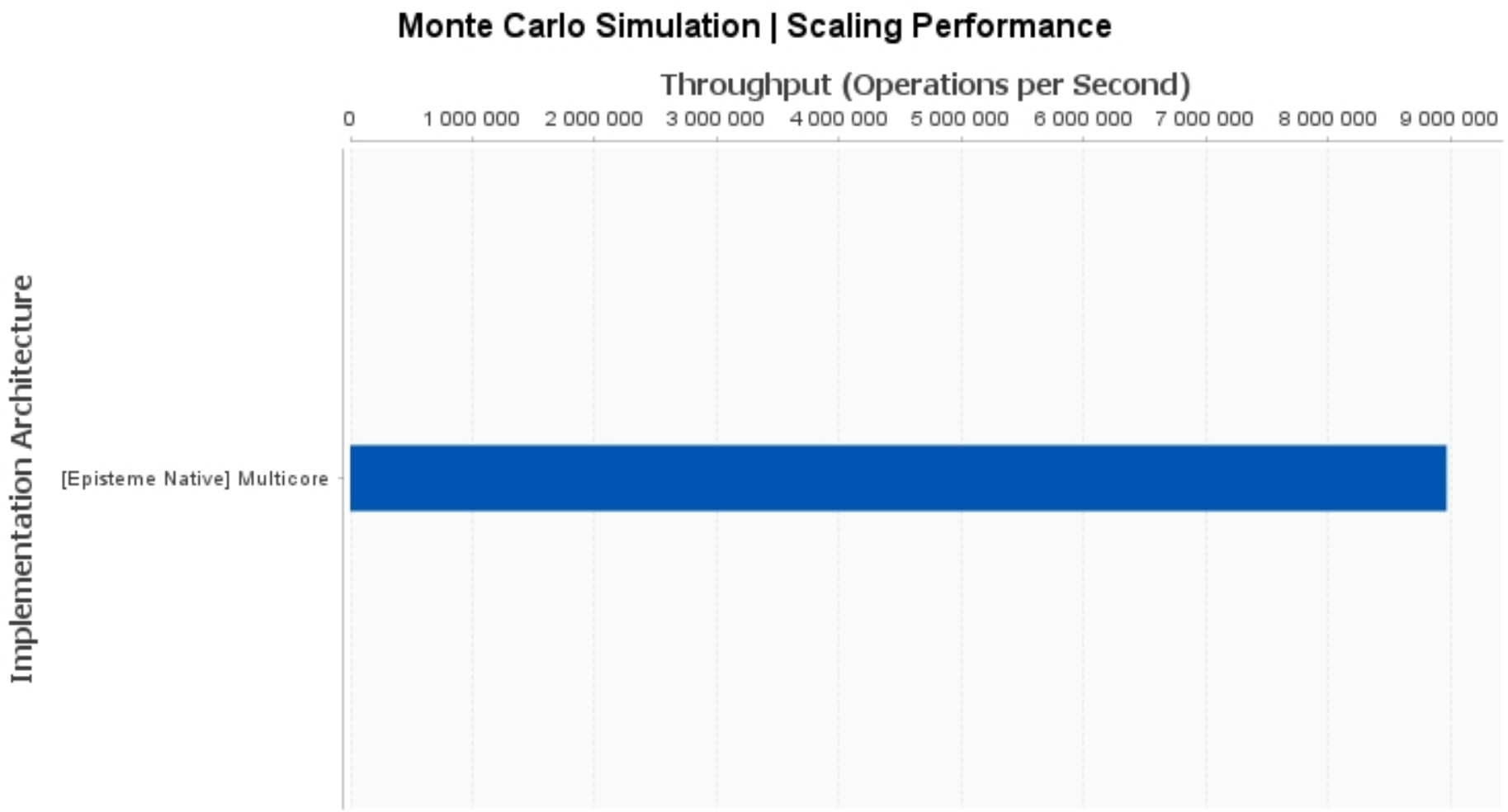
Molecular Dynamics



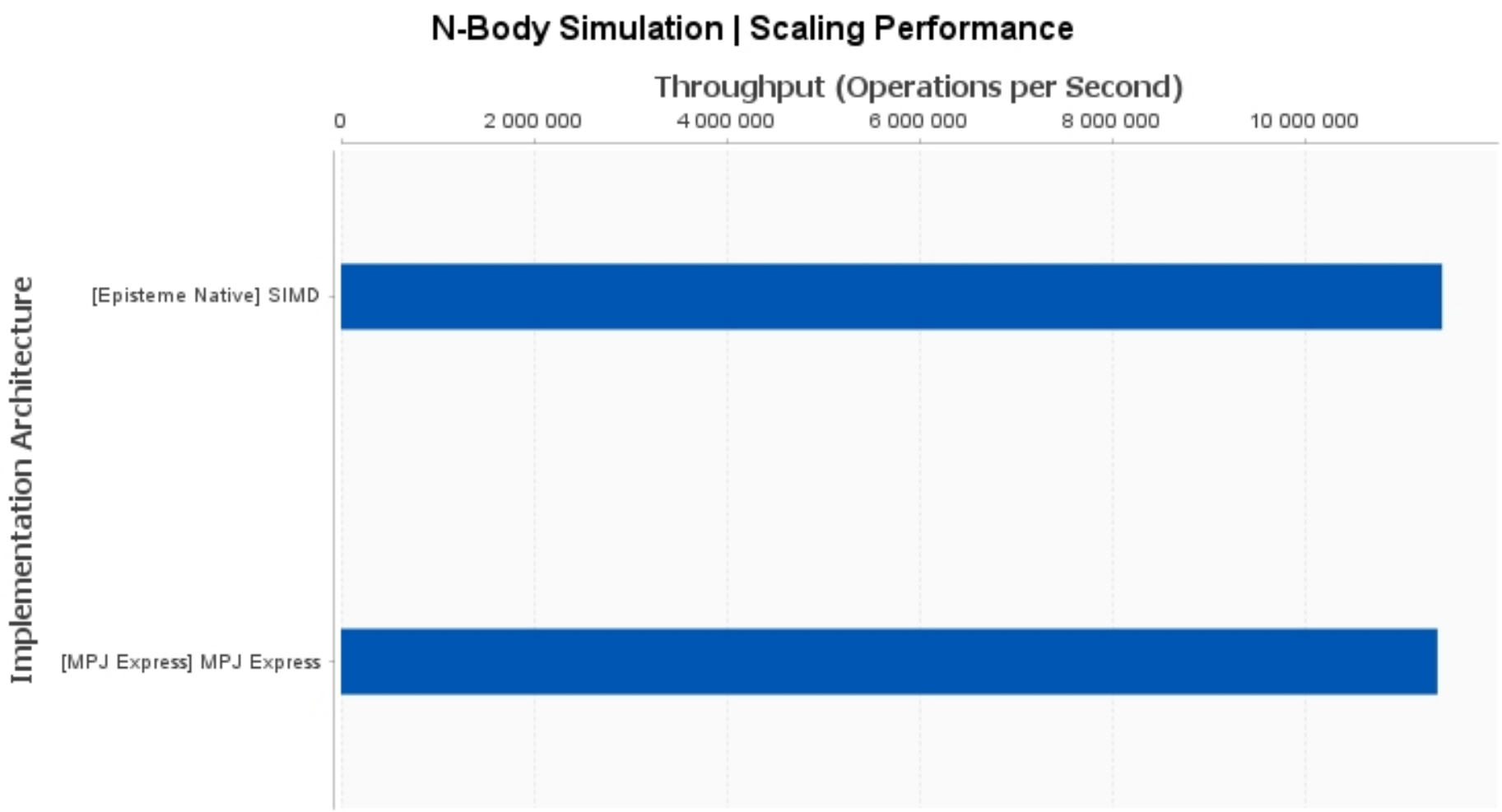
Monte Carlo Simulation



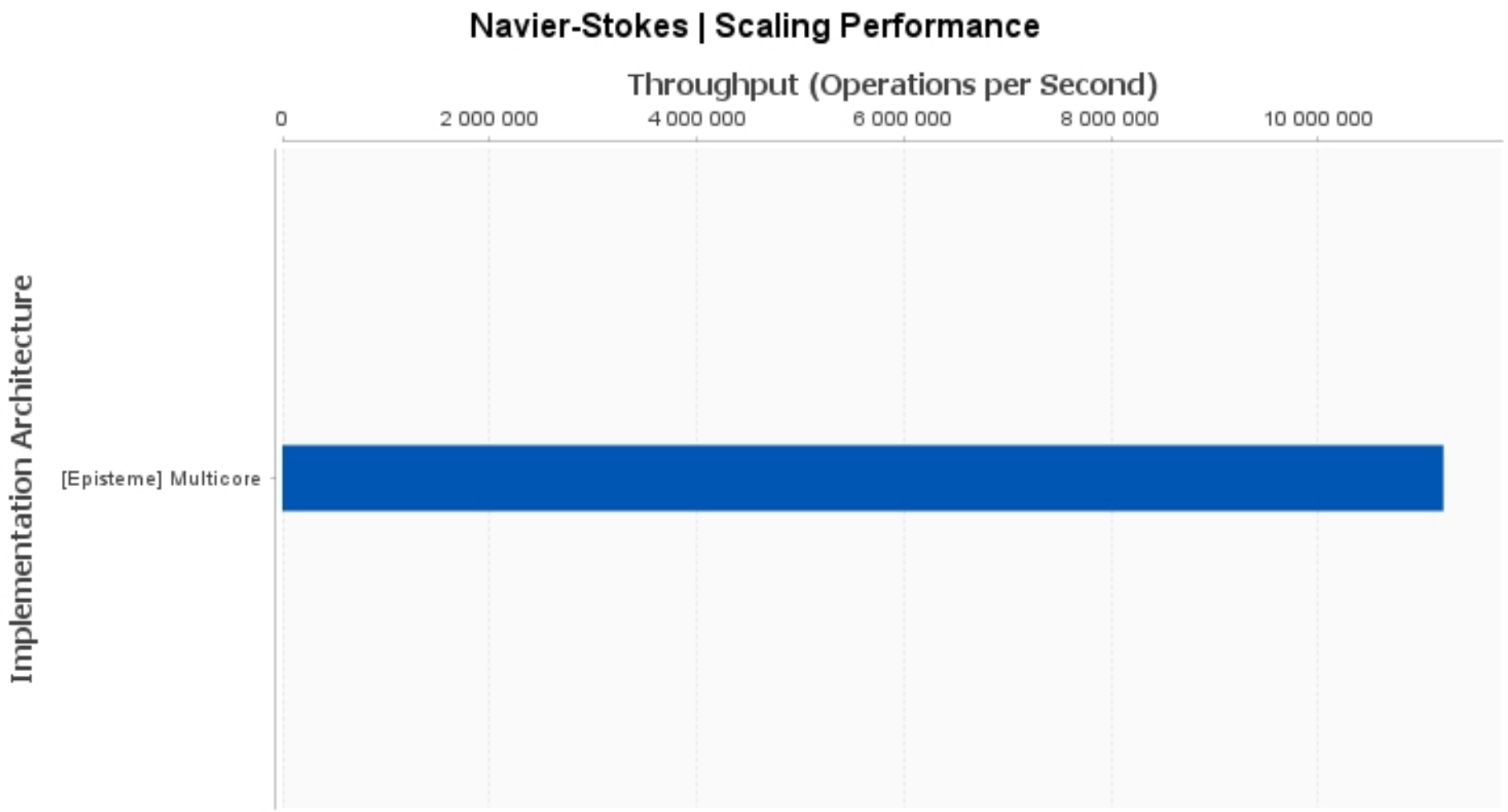
N-Body Simulation



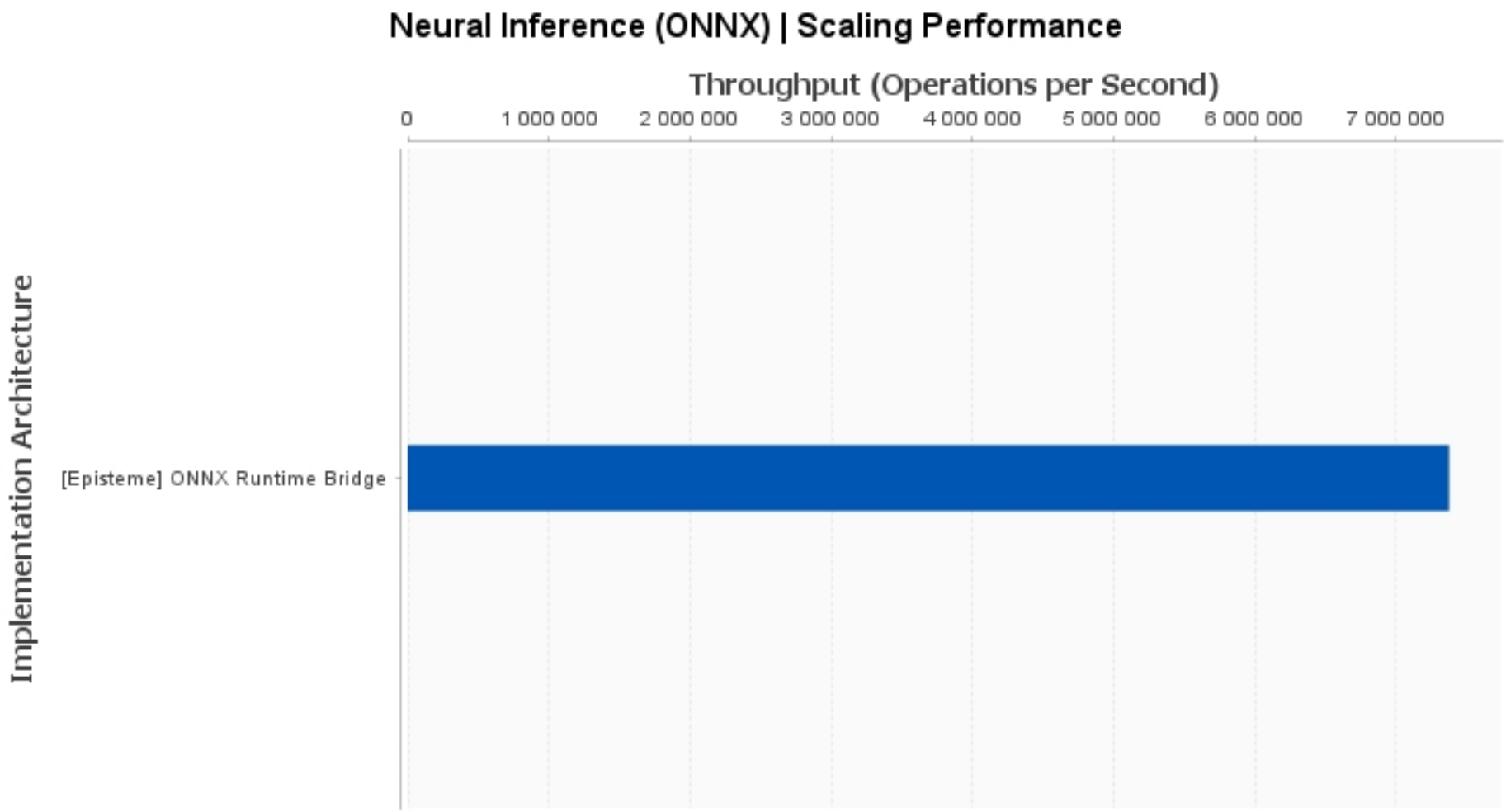
Navier-Stokes



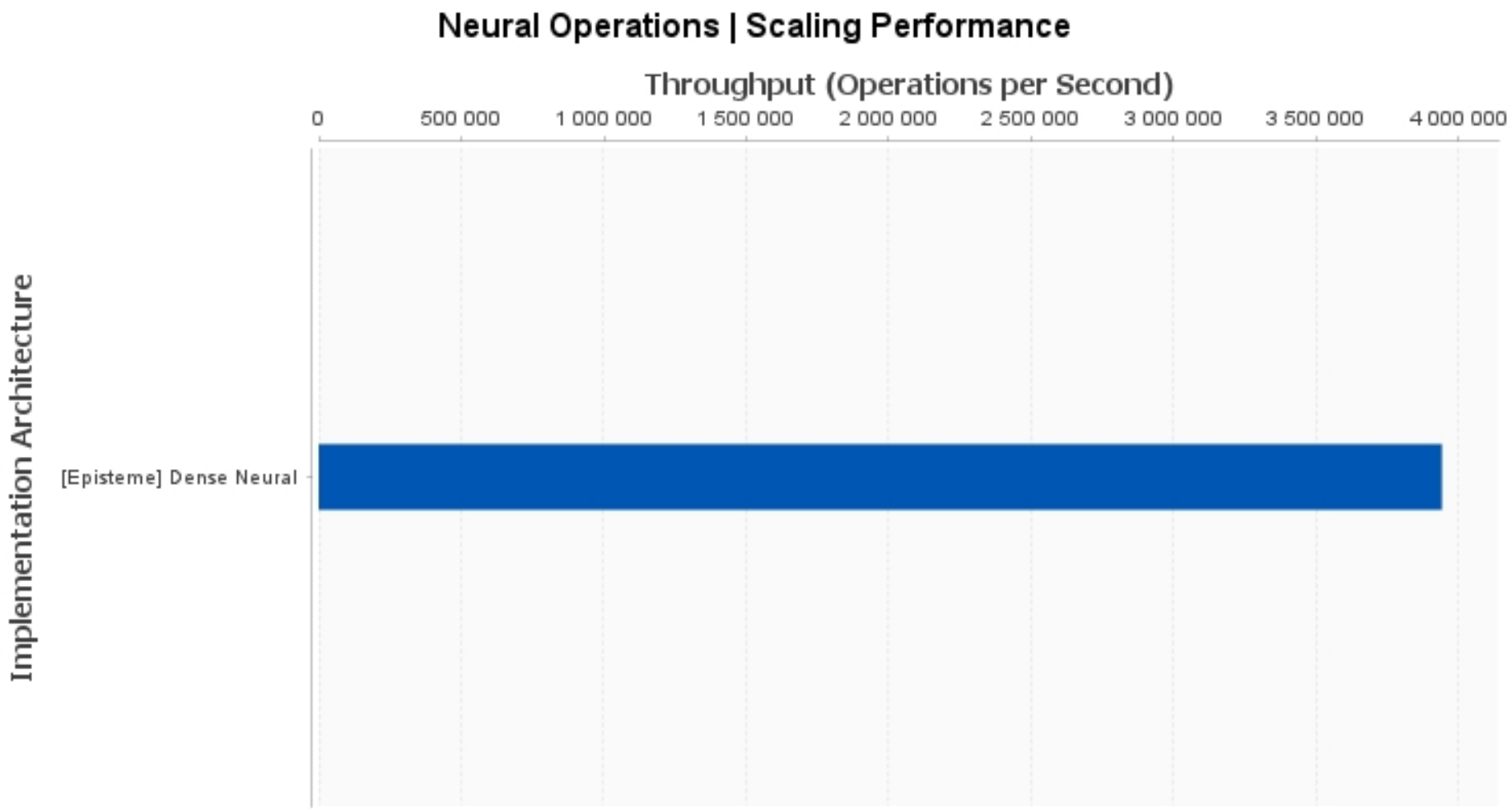
Neural Inference (ONNX)



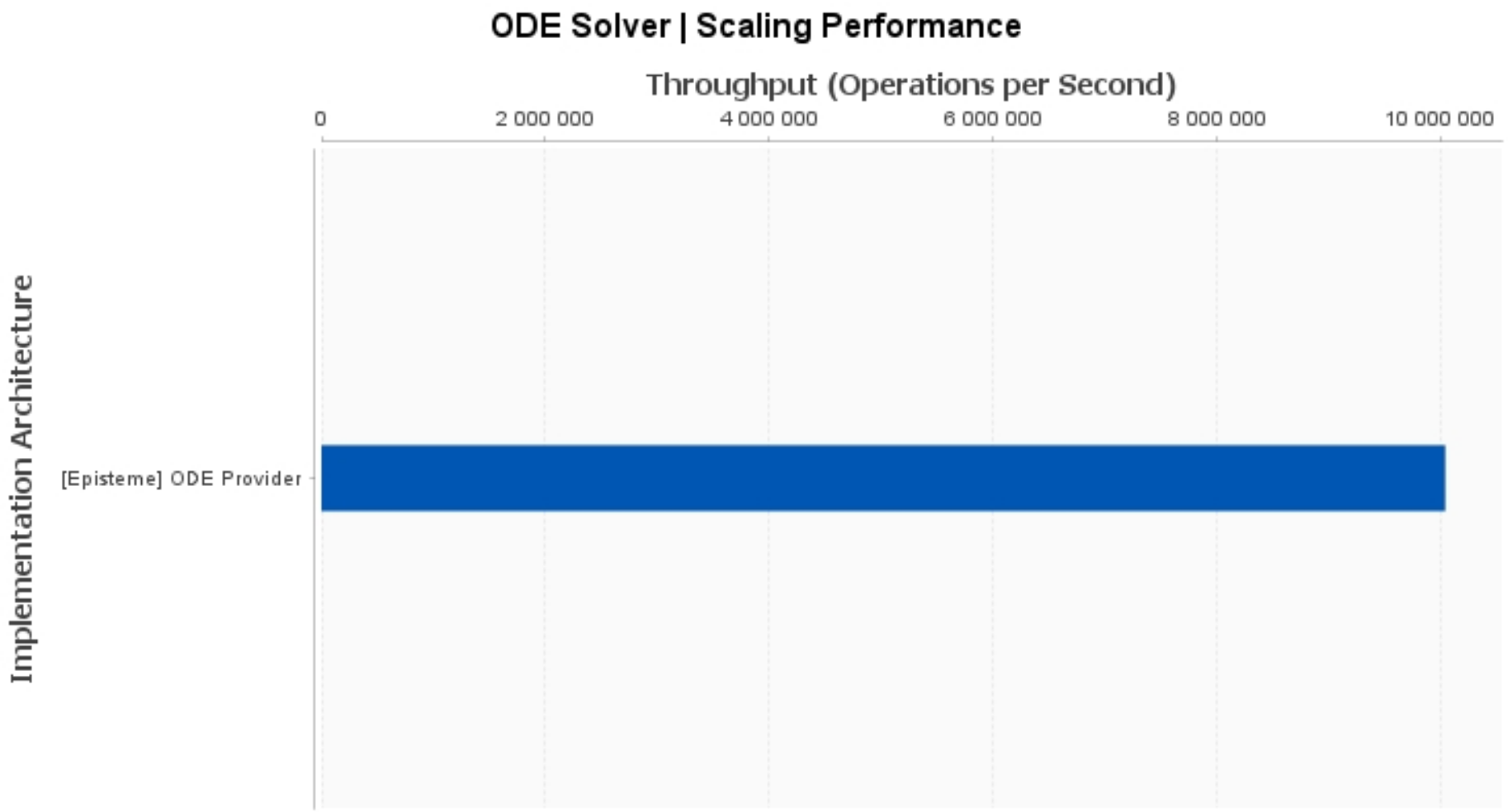
Neural Operations


Neural Inference (ONNX) | Scaling Performance
Throughput (Operations per Second)
0
1 000 000
2 000 000
3 000 000
4 000 000
5 000 000
6 000 000
7 000 000
Implementation Architecture
[Episteme] ONNX Runtime Bridge

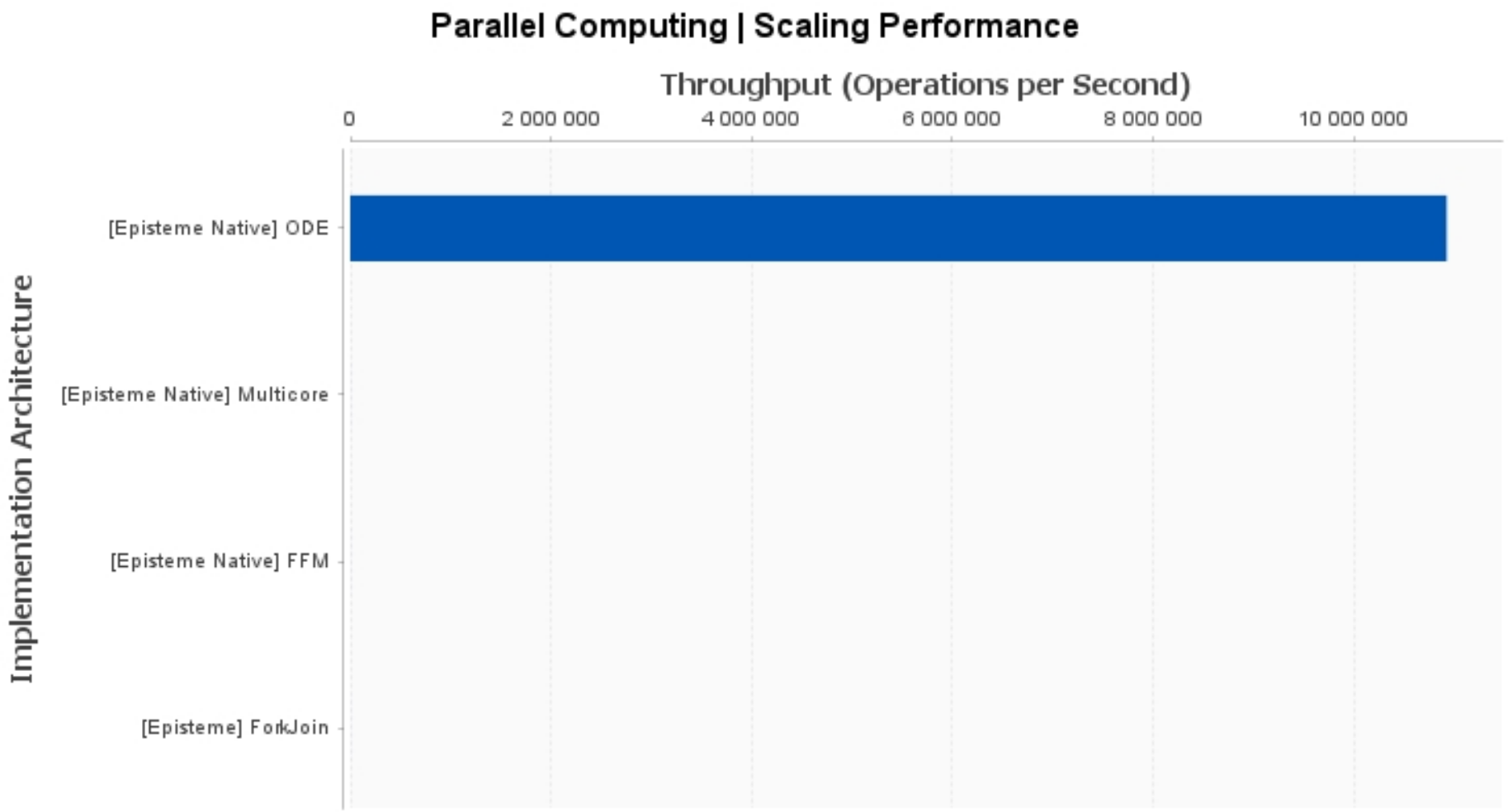
ODE Solver



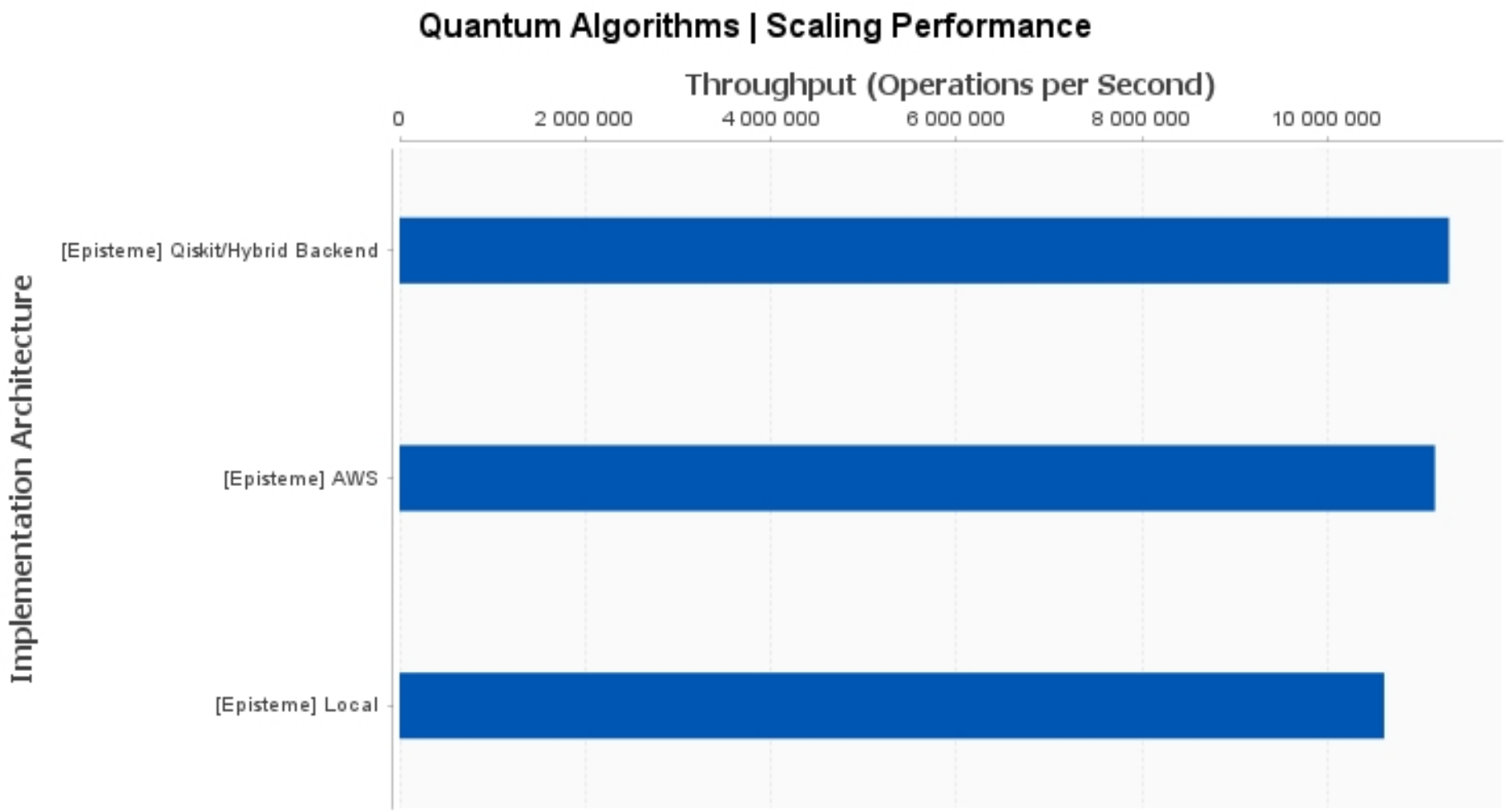
Parallel Computing



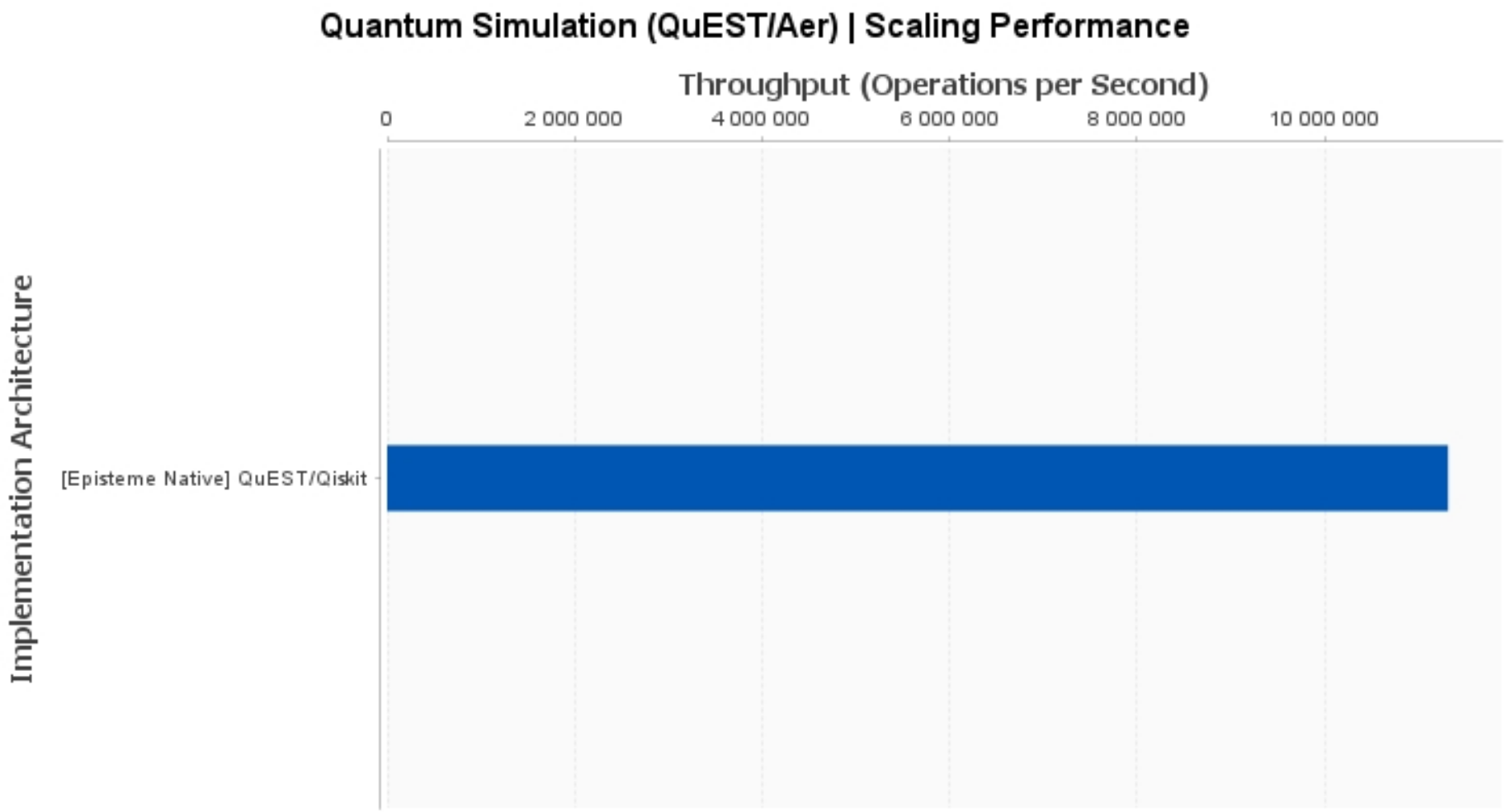
Quantum Algorithms



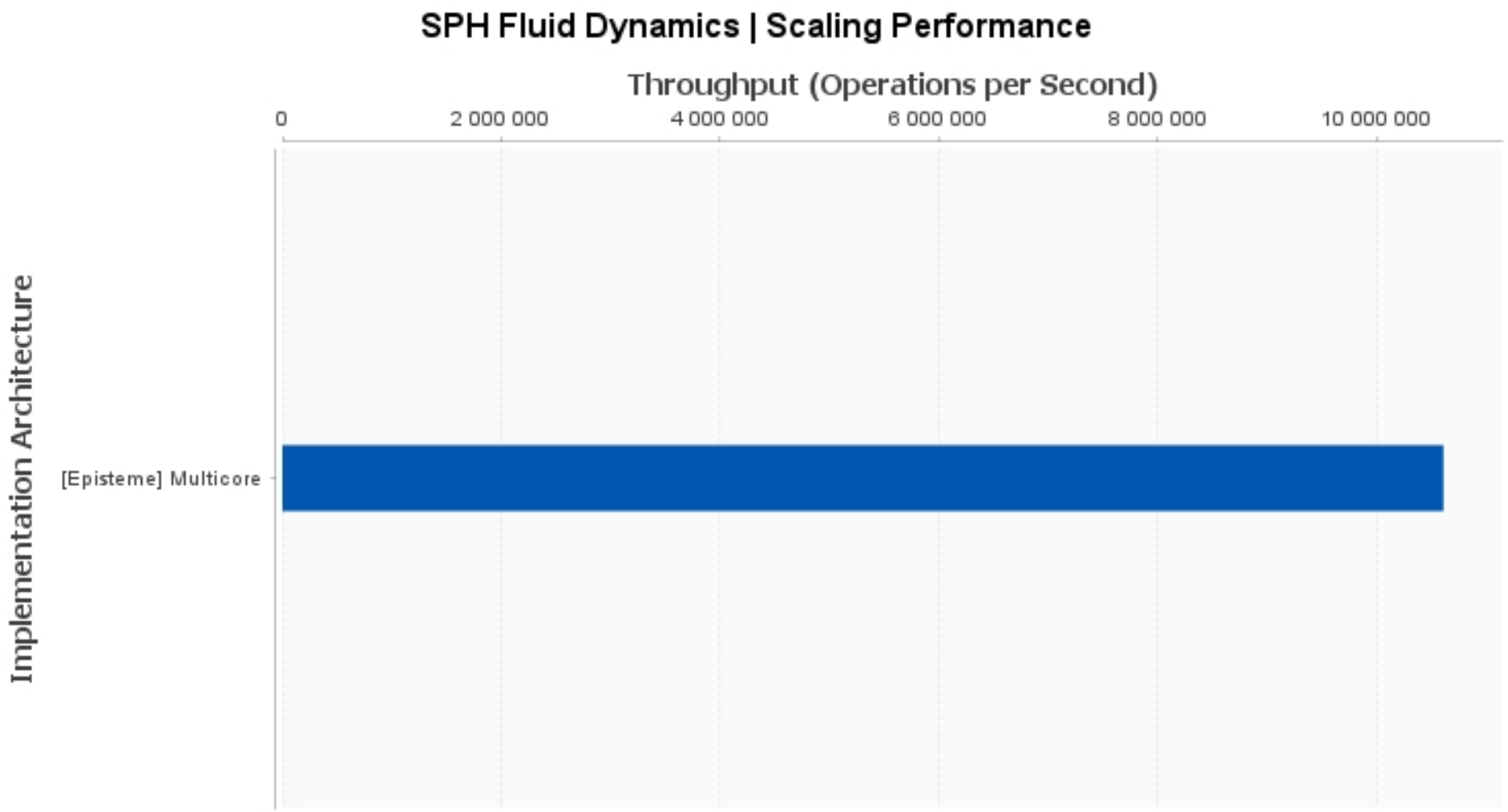
Quantum Simulation (QuEST/Aer)



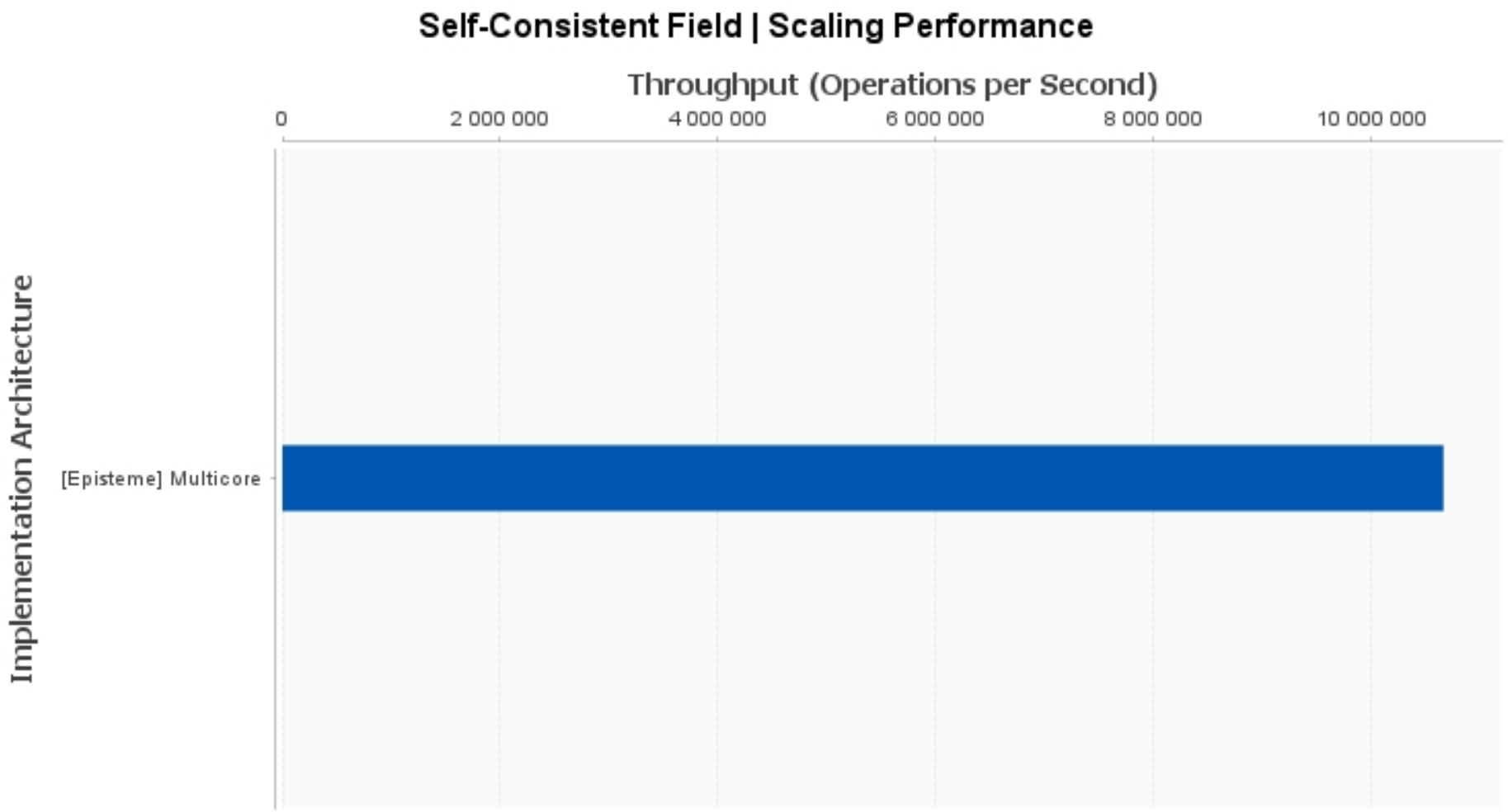
SPH Fluid Dynamics



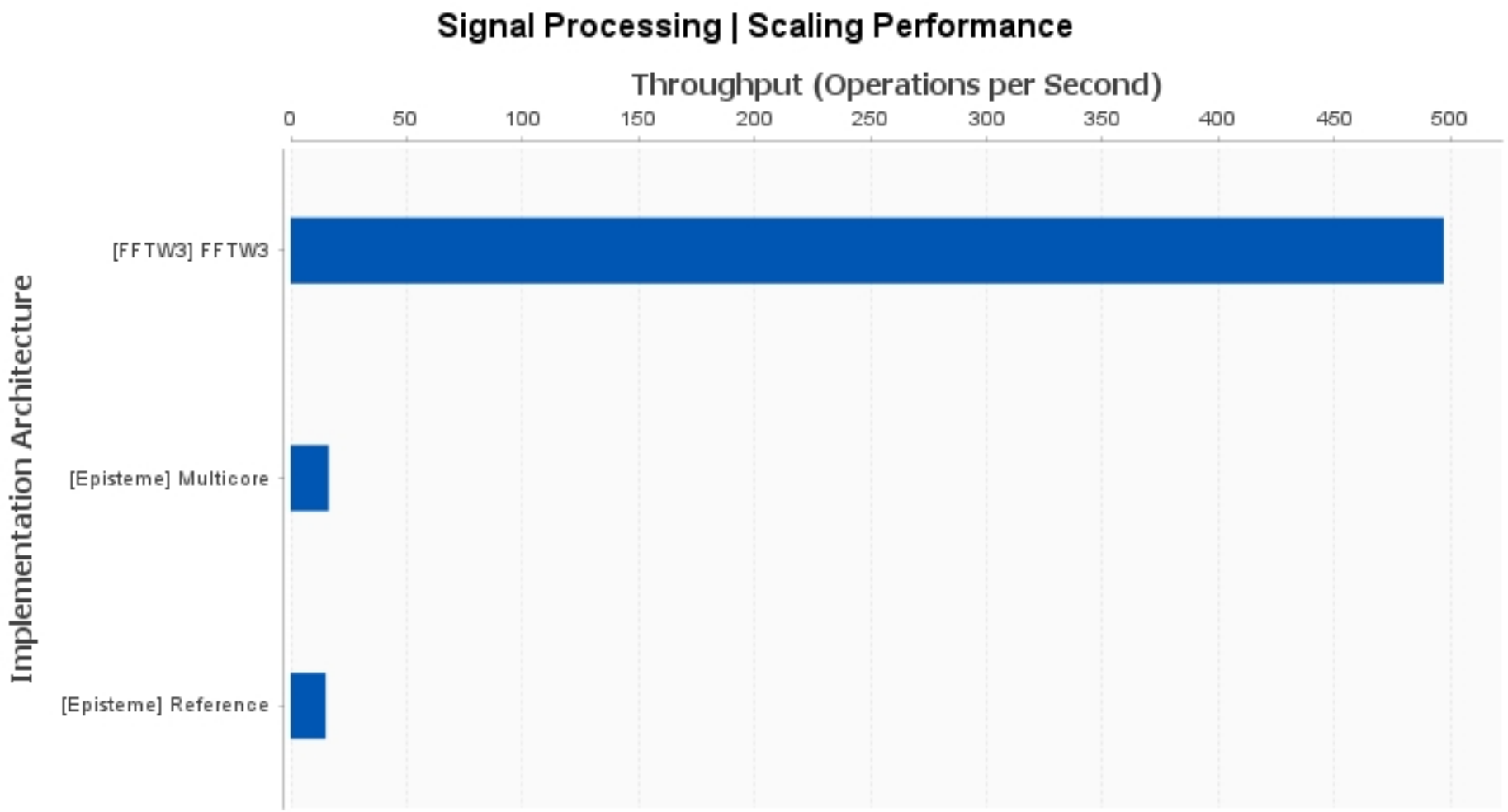
Self-Consistent Field



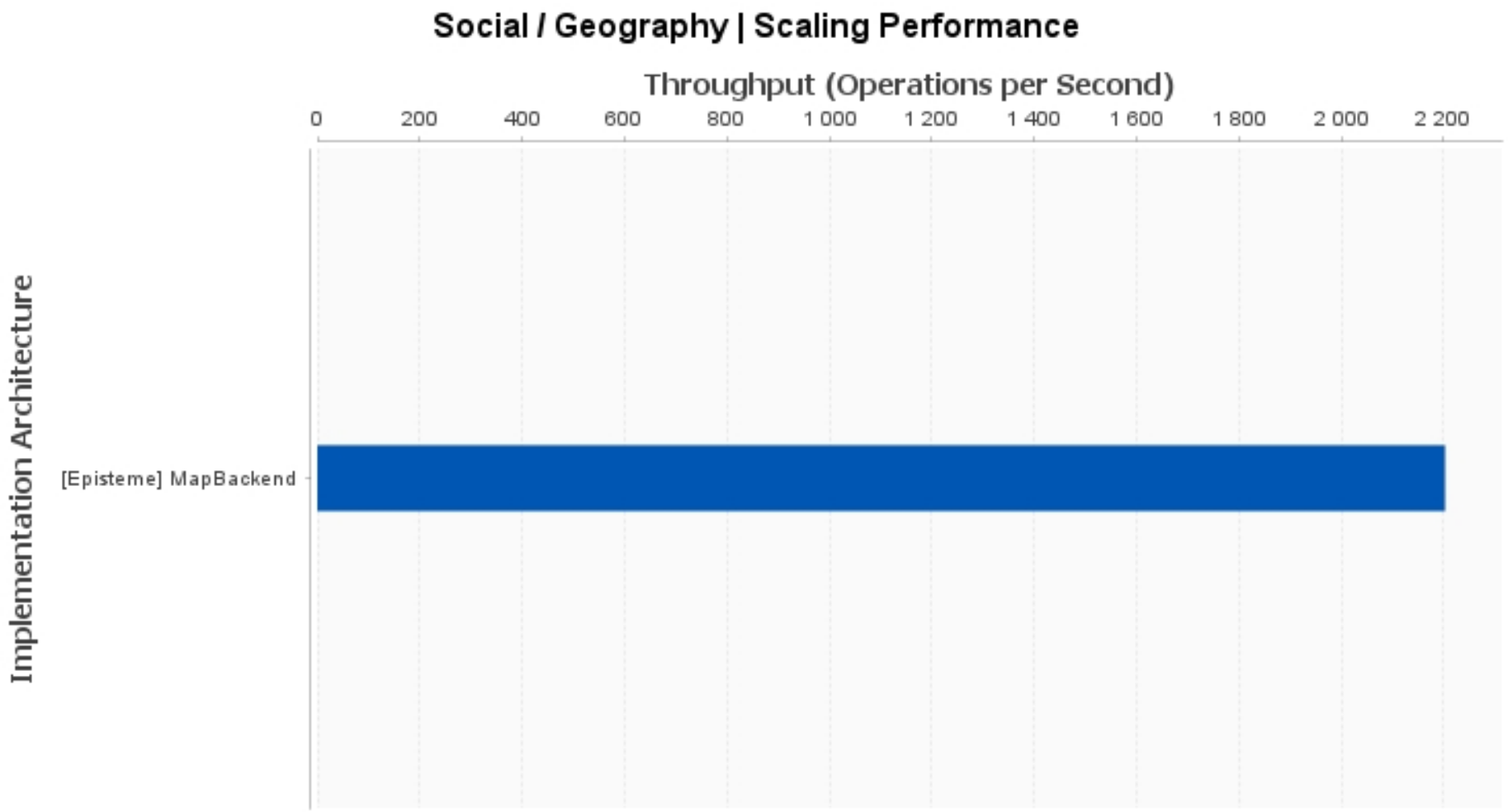
Signal Processing



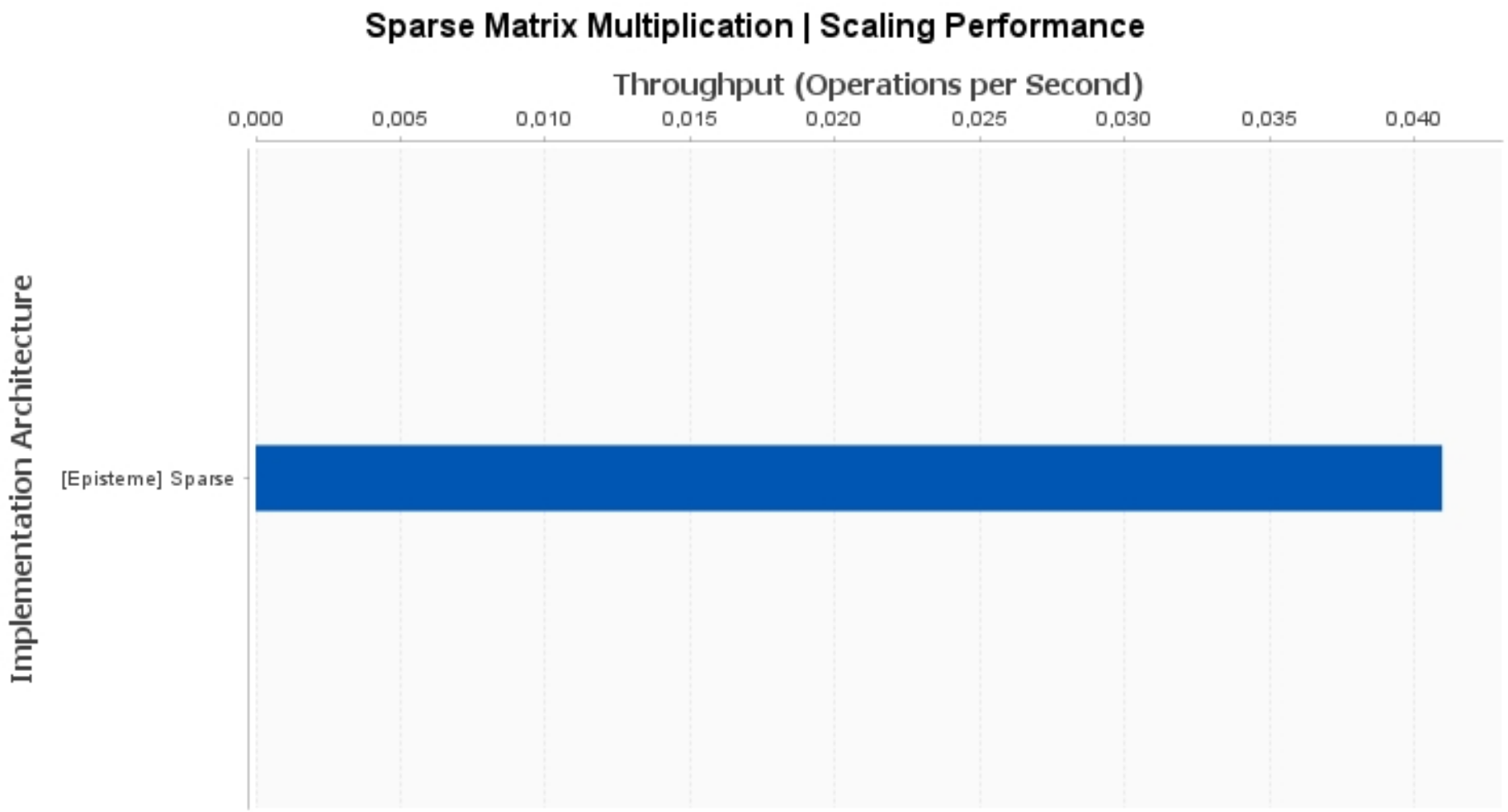
Social / Geography



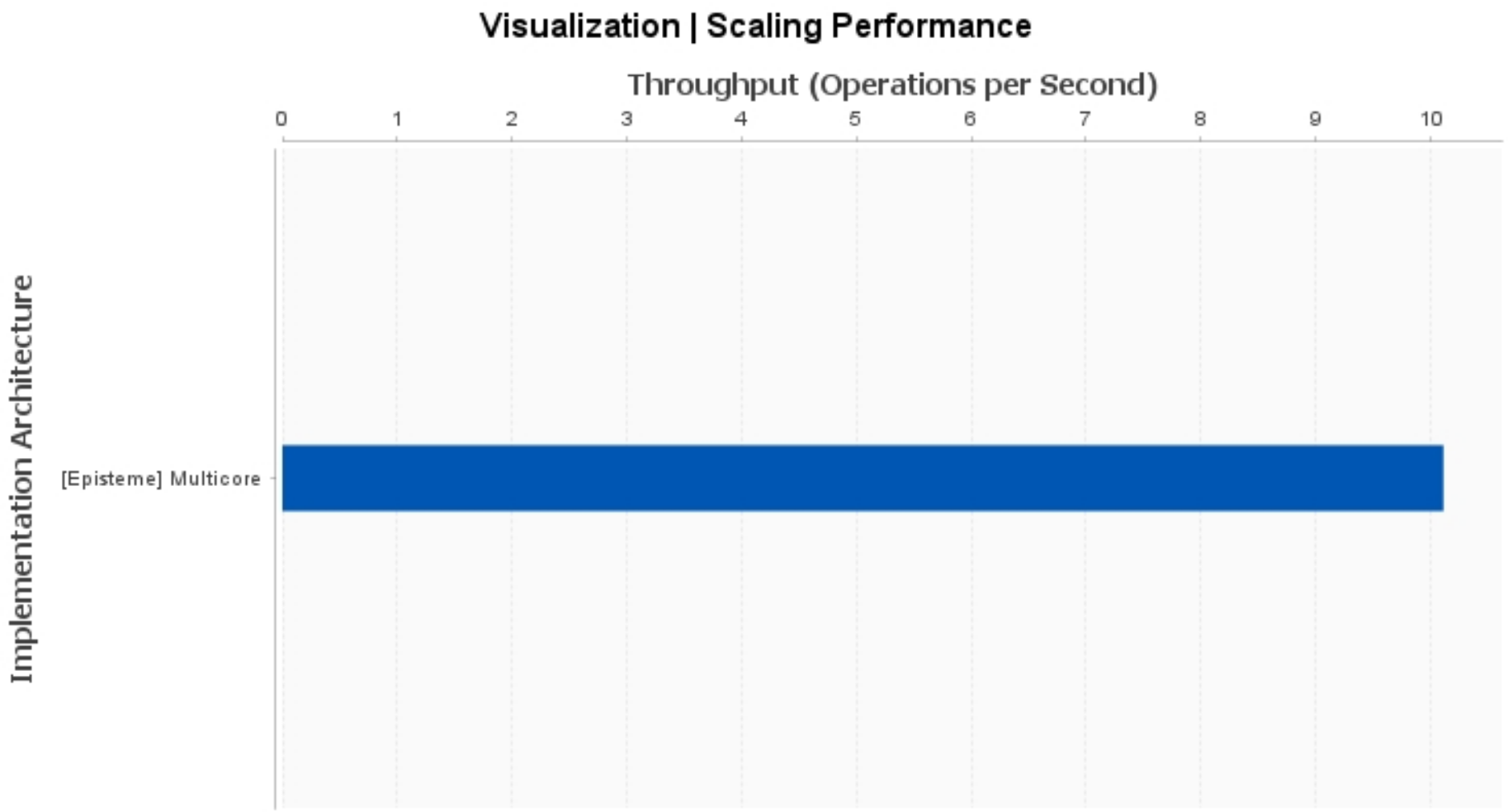
Sparse Matrix Multiplication



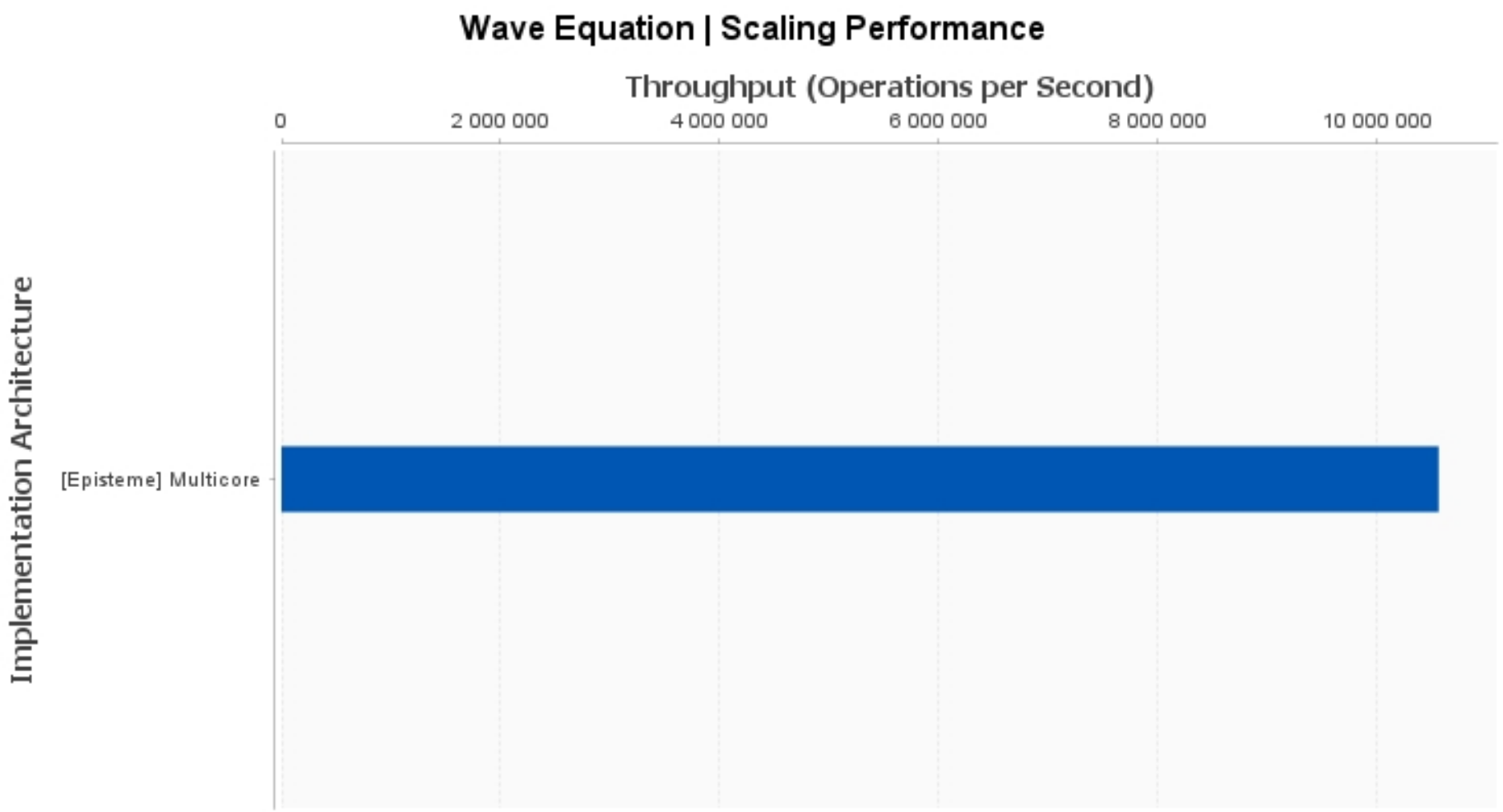
Visualization

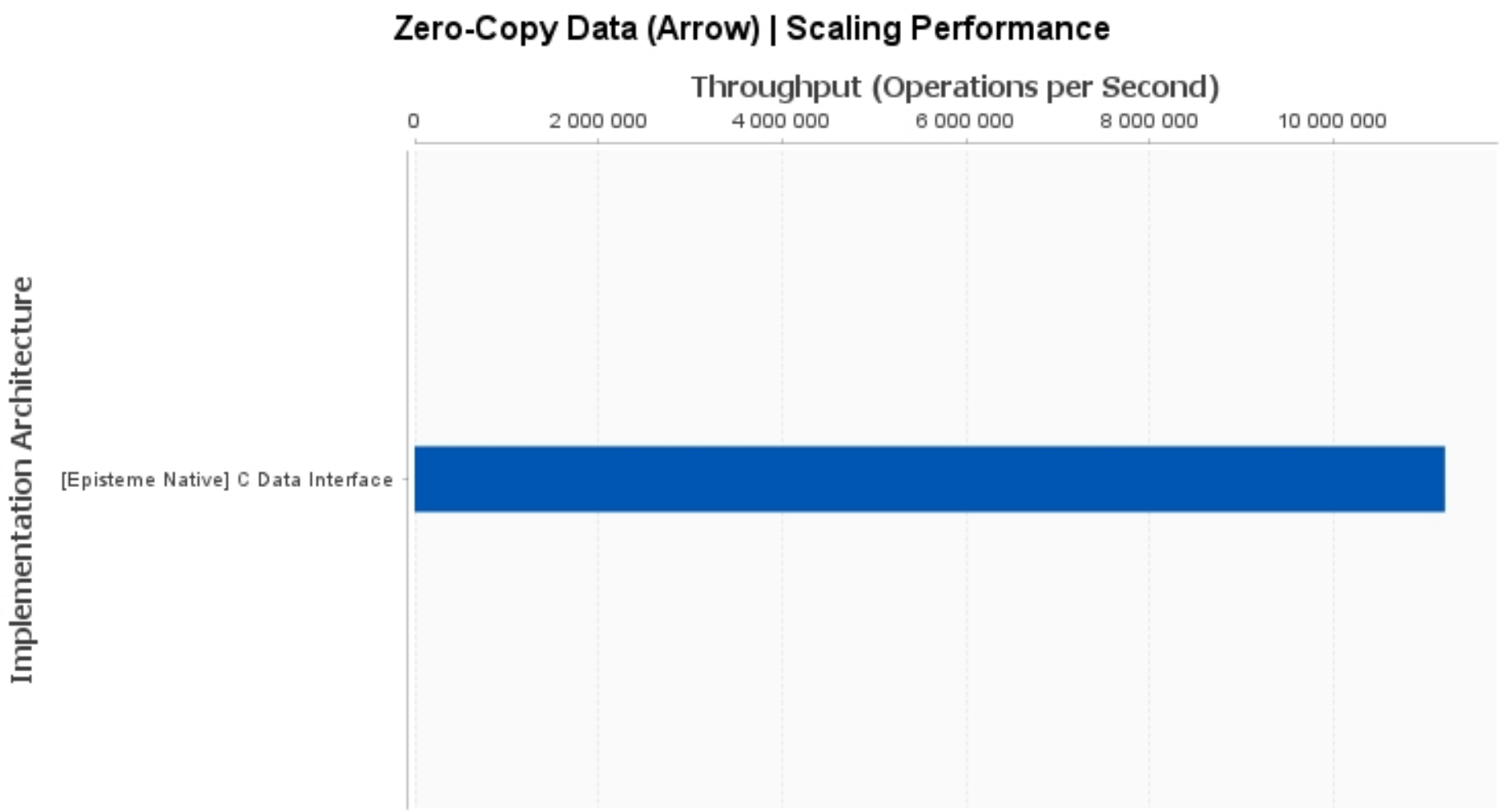


Wave Equation



Zero-Copy Data (Arrow)


Wave Equation | Scaling Performance
Throughput (Operations per Second)
0
2 000 000
4 000 000
6 000 000
8 000 000
10 000 000
Implementation Architecture
[Episteme] Multicore


Zero-Copy Data (Arrow) | Scaling Performance
Throughput (Operations per Second)
0
2 000 000
4 000 000
6 000 000
8 000 000
10 000 000
Implementation Architecture
[Episteme Native] C Data Interface